Wave State Propagation for Irregular-Time Neural Dynamics Modeling

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Abstract

Sequence learning from irregular observations requires models to distinguish elapsed physical time from observation order. We introduce Wave State Propagation (WSP), a wave-parameterized neural approach that combines content-conditioned injection with persistent state evolution over measured intervals. Derived from Wave Function Backpropagation, WSP uses amplitude A, latent wavenumber k, and phase θ to construct wave states, while angular frequency ω and damping $\exp(-\gamma\Delta t/T)$ govern closed-form propagation at a training-derived time scale T. An exact paired-real implementation retains both quadratures, and a zero-time displacement readout links latent evolution to prediction. A parameter-matched benchmark compares persistent WSP with five temporal baselines on four datasets using ten final training seeds. On HalfCheetah, WSP reduces test normalized root-mean-square error by 27.2% versus LRU and 35.3% versus a time-conditioned Transformer. Shuffled timing degrades WSP on time-relevant tasks but has negligible effect on the time-irrelevant control. Rankings remain task-dependent: WSP's oscillator advantage over Transformer is unresolved, while LRU is slightly better on PhysioNet. At validation accuracy matched within 1% on PhysioNet, batch-256 WSP inference uses 50.9% less peak allocated memory than Transformer but is 22.9 times slower on an NVIDIA B200. These findings support WSP as a viable approach to sequence prediction involving continuous physical time, with application-dependent accuracy and memory–throughput trade-offs.

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1. Introduction

Sequence order and physical time are not interchangeable. Positional encoding indicates whether one observation precedes another, whereas an elapsed interval specifies how long the underlying system evolves. Two records can therefore have identical content and order but imply different transitions when their sampling intervals differ. This distinction is central to asynchronous sensing, event streams, robotics, and continuous control. A model that treats Δt only as another feature must learn the corresponding flow map indirectly.

Continuous-time neural models address this problem in several ways. Time-dependent decay modifies recurrent states according to observation gaps [1]; latent ordinary differential equations and neural controlled differential equations evolve hidden variables between observations [2, 3]; and ContiFormer combines continuous trajectories with attention [4]. Structured state-space models (SSMs) offer a complementary closed-form route: their diagonal or diagonalizable continuous generators support stable recurrence and parallel scans [5–8]. Selective SSMs further make transition parameters input-dependent [9]. WSP builds on this continuous-time SSM foundation: its homogeneous physical-time flow is complex diagonal, while its architecture combines WFB-derived content phase and amplitude with measured-interval propagation and a zero-time displacement construction. The contribution is this wave-parameterized design and its empirical assessment for physical-time sequence prediction.

Wave Function Backpropagation (WFB) introduced a wave response with learnable amplitude A, wavenumber $\mathbf{k}$, angular frequency ω, and phase θ [10]. For input $\mathbf{x}$ and elapsed time Δt, WFB uses the phase $\mathbf{k}^\top \mathbf{x} - \omega \Delta t - \theta$. Its feed-forward evaluation recomputes that response at each sample; it does not carry a wave state through an ordered observation sequence.

This paper introduces *Wave State Propagation* (WSP). The evaluated physical-time operator is $\exp(-\gamma \Delta t/T) \exp(-\mathrm{i}\omega \Delta t/T)$, where T is a training-derived time scale. Wavenumber k and phase θ determine the content-dependent injection phase, whereas ω and γ determine rotation and attenuation over measured time. The current observation initializes or updates

the wave state, which is then propagated over its associated interval. Paired real-valued cosine and sine channels implement the complex recurrence exactly. We study both a transition-local specialization, which initializes a wave for each supervised transition, and a persistent specialization, which carries state through ordered windows. Keeping these variants separate is necessary because individual-transition prediction does not by itself test across-observation persistence.

The primary empirical study evaluates the persistent recurrence on a damped oscillator, HalfCheetah dynamics, a PhysioNet sequence-prediction adaptation, and a synthetic control whose transition target does not depend on elapsed time. A shared validation-only search precedes ten-seed test comparisons against temporal recurrent models and a zero-time-constrained Transformer. A separately configured component study examines carry, damping, quadratures, conditioning, and the zero-time constraint; development diagnostics compare additional state-space and transition-local models. These protocols are reported separately rather than pooled into a single performance claim.

The central question is whether this wave-parameterized architecture offers a useful approach to sequence learning when measured elapsed time informs the underlying dynamics. We assess that question through predictive accuracy, interventions on the time input, and measured computational costs. The goal is to establish WSP's viability and clarify its operating trade-offs, rather than to assume that the availability of Δt makes it preferable to every alternative.

The contributions of this work are threefold:

1. We introduce a wave-parameterized neural architecture that combines WFB-derived content injection, persistent latent states, and explicit propagation over measured physical intervals.
2. We specify transition-local and persistent specializations with an exact paired-real implementation and a zero-time displacement readout. We characterize the fixed-generator identity, homogeneous stability, and affine-composition properties within the continuous-time SSM family.
3. We evaluate predictive accuracy, reliance on elapsed time, open-loop replay, and accuracy-matched inference costs using a parameter-matched ten-seed benchmark, complemented by independent component and development diagnostics. The results establish useful physical-time prediction behavior and identify application-dependent accuracy and

memory–throughput trade-offs.

The remainder of the paper is organized as follows. Section 2 reviews continuous-time sequence models, neural SSMs, and world models. Section 3 summarizes the wave-function foundation inherited from WFB. Section 4 develops WSP and its mathematical properties. Section 5 presents the experimental protocol and results. Section 6 concludes the paper and identifies directions for extending WSP to deeper sequence and control architectures.

2. Related Work

2.1. Irregular and Continuous-Time Sequence Learning

Conventional recurrent and attention-based models operate primarily over an ordinal sequence index. Irregularly sampled data require an additional model of the duration between observations. GRU-D introduces interval-dependent decay to handle missing multivariate measurements [1]. Latent ODEs instead infer a latent initial state and evolve it with a neural ordinary differential equation [2, 11], while GRU-ODE-Bayes and liquid time-constant networks combine recurrent updates with continuous flows [12, 13]. Phased LSTM gates updates with a learned oscillatory clock for sparse event streams [14]. Continuous Recurrent Units (CRUs) use continuous-discrete Kalman updates and closed-form latent dynamics; the efficient f-CRU variant restricts the transition parameterization [15]. UnICORNN derives a recurrent architecture from a structure-preserving discretization of coupled oscillator dynamics [16]. These studies establish oscillatory and continuous-time dynamics as useful foundations for sequence learning. Neural controlled differential equations use an interpolated observation path to drive continuous hidden dynamics [3]. WSP builds on these foundations by combining content-dependent amplitude and phase with a closed-form wave propagation operator whose frequency and damping specify rotation and attenuation over each measured interval. The evaluation includes a preprocessed PhysioNet sequence-prediction task, distinct from clinical outcome prediction, observation-mask-aware imputation, and calibrated stochastic latent inference.

2.2. Structured State-Space Models

S4, diagonal state-space layers, and S5 show that continuous linear generators can be discretized into efficient long-sequence recurrences [5–7]. LRU

obtains related benefits with a carefully parameterized diagonal complex recurrence, and Mamba makes SSM parameters input-selective [8, 9]. For a fixed interval and conditioning context, WSP's homogeneous multiplier is a diagonal complex SSM. WSP differs in how WFB variables initialize the content phase, how measured Δt directly discretizes the flow, and how the transition-local model reads the change from a zero-time wave. The primary sequence benchmark includes a parameter-matched LRU adaptation. S5 supplies related mathematical context rather than a native benchmark implementation, while independently tuned diagonal-SSM diagnostics are reported separately. Related Koopman approaches also seek latent coordinates with simple linear evolution, but generally learn the embedding rather than prescribe a WFB phase parameterization [17].

2.3. Continuous-Time Attention and World Models

Transformers use attention to model long-range dependencies [18]. Positional encoding alone does not specify elapsed physical time. ContiFormer extends attention to irregular continuous-time data by combining neural ODE trajectories with attention integration [4]. WSP shares the objective of modeling continuous evolution but replaces numerical vector-field integration with an analytic, state-conditioned phase propagation.

Learned world models provide another route to continuous-control dynamics. TD-MPC2 learns a decoder-free latent model used for local trajectory optimization [19]. DCWM introduces stochastic discrete codebook states and combines them with decision-time planning in DC-MPC [20]. Their native objectives include latent consistency, reward or value prediction, policies, and planning. Their control objectives differ from the supervised displacement prediction studied here. Our empirical claims concern sequence prediction and recorded-action open-loop replay, not the native closed-loop return of DC-MPC or TD-MPC2.

2.4. Periodic Neural Representations and WFB

Periodic activations provide a compact basis for continuously varying and high-frequency signals. Sinusoidal representation networks, for example, show that periodic nonlinearities can encode detailed signals and their derivatives [21]. WFB extends this perspective by jointly learning amplitude, state-dependent wavenumber, temporal angular frequency, and phase, with Δt appearing directly in the wave phase term [10]. WSP contributes two

physical-time uses of this basis: a zero-time wave increment for individual transitions and a compositional carry recurrence for ordered sequences.

3. Mathematical Formulation of the Wave Function

WSP is an extension of Wave Function Backpropagation (WFB) [10], rather than an unrelated continuous-time architecture. For self-contained notation, Eqs. (1) and (2) restate the complex WFB wave response and its practical real-valued activation, respectively. They correspond to Eqs. (2) and (5), respectively, in the cited WFB paper, with channel index m made explicit here:

$$\psi_m(\mathbf{x}, \Delta t) = A_m \exp\Big[\mathrm{i}\left(\mathbf{k}_m^\top \mathbf{x} - \omega_m \Delta t - \theta_m\right)\Big], \tag{1}$$

whose real-valued activation is

$$f_m(\mathbf{x}, \Delta t) = A_m \cos\Big(\mathbf{k}_m^\top \mathbf{x} - \omega_m \Delta t - \theta_m\Big). \tag{2}$$

Equation (2) takes the real part only because the original pointwise WFB activation must return a real feature. WSP does not discard the imaginary component during propagation: it retains both quadratures and implements them as paired real tensors. The evaluated WSP code uses $+\theta$ in its injection phase; this is the harmless reparameterization $\theta_{\mathrm{WSP}} = -\theta_{\mathrm{WFB}}$ of the WFB convention above. Amplitude A_m controls response magnitude, wavenumber $\mathbf{k}_m$ controls variation with the state or spatial input, angular frequency ω_m controls phase change per unit elapsed time, and θ_m is a phase offset. WFB derives the task-loss gradients of these parameters explicitly. For compactness, their ordinary gradient-descent update can be summarized as

$$q \leftarrow q - \eta \frac{\partial \mathcal{L}_{\mathrm{task}}}{\partial q}, \qquad q \in \{A_m, \mathbf{k}_m, \omega_m, \theta_m\}. \tag{3}$$

Equation (3) is a standard optimization summary, not an additional wave equation quoted verbatim from WFB. The WFB proof of concept applies Eq. (1) as an interval-conditioned representation inside an FFN. It therefore changes the activation at each observation but does not preserve and propagate a wave state between successive observations.

4. Wave State Propagation

This section gives the formulation used by the reported implementations. WSP contains two specializations: a transition-local wave increment for individual-transition diagnostics and a persistent carry recurrence used by the primary ordered-window benchmark. Both use the same physical-time operator, but only the second carries a state across observations.

4.1. Physical-Time Generator

For transition i, observation $\mathbf{x}_i$ is recorded at time τ_i and evolves to $\mathbf{x}_{i+1}$ over

$$\Delta t_i = \tau_{i+1} - \tau_i, \qquad \widetilde{\Delta t}_i = \frac{\Delta t_i}{T}, \tag{4}$$

where T is the largest training interval. Ordinal position may be encoded by a surrounding sequence model, but the evaluated WSP dynamics has no independent positional propagation coordinate.

For wave channel m and layer ℓ, the homogeneous continuous dynamics over one interval are expressed in normalized time $u = t/T$:

$$\frac{\mathrm{d}\psi_m^{(\ell)}}{\mathrm{d}u} = \left(-\gamma_{i,m}^{(\ell)} - \mathrm{i}\omega_{i,m}^{(\ell)}\right) \psi_m^{(\ell)}, \tag{5}$$

with the state-conditioned parameters held fixed during that interval. Equivalently, the physical-time generator is $(-\gamma_{i,m}^{(\ell)} - \mathrm{i}\omega_{i,m}^{(\ell)})/T$. The exact flow for normalized duration $u \geq 0$ is

$$\mathcal{P}_i^{(\ell)}(u) = \mathrm{diag}\Big[\exp\Big(-\boldsymbol{\gamma}_i^{(\ell)} u\Big) \odot \exp\Big(-\mathrm{i}\boldsymbol{\omega}_i^{(\ell)} u\Big)\Big]. \tag{6}$$

Thus ω controls physical-time rotation and γ controls attenuation. The inherited wavenumber k does not appear in Eq. (6); in the evaluated models it controls the latent content-injection phase. This distinction corrects a possible but inaccurate interpretation of k as an additional token-position transition.

Equation (5) also makes the relationship to neural SSMs explicit. Its generator is a diagonal complex state matrix, so the homogeneous WSP flow is mathematically a complex diagonal continuous-time SSM [6–8]. WSP's contribution lies in its WFB-derived content parameterization, measured-time discretization, zero-time displacement readout, and persistent injection rule.

4.2. State-Conditioned Wave Parameters

Let $\mathbf{h}_i^{(\ell-1)} \in \mathbb{R}^{d_h}$ be the hidden content and $\mathbf{c}_i^{(\ell)} = \mathrm{LN}(\mathbf{h}_i^{(\ell-1)})$. A rank-16 conditioner produces five channel-wise shifts,

$$[\boldsymbol{\delta}_A, \boldsymbol{\delta}_k, \boldsymbol{\delta}_\omega, \boldsymbol{\delta}_\gamma, \boldsymbol{\delta}_\theta]_i^{(\ell)} = \mathrm{split}_5\Big(\mathbf{W}_{\mathrm{up}}^{(\ell)} \mathbf{W}_{\mathrm{down}}^{(\ell)} \mathbf{c}_i^{(\ell)}\Big). \tag{7}$$

The evaluated parameterization is

$$\begin{aligned} \mathbf{A}_i^{(\ell)} &= \mathrm{softplus}(\mathbf{a}^{(\ell)} + \boldsymbol{\delta}_A), & \mathbf{k}_i^{(\ell)} &= \mathrm{softplus}(\boldsymbol{\kappa}^{(\ell)} + \boldsymbol{\delta}_k), \\ \boldsymbol{\omega}_i^{(\ell)} &= \omega_{\max}\sigma(\mathbf{o}^{(\ell)} + \boldsymbol{\delta}_\omega), & \boldsymbol{\gamma}_i^{(\ell)} &= \gamma_{\max}\sigma(\mathbf{r}^{(\ell)} + \boldsymbol{\delta}_\gamma), \\ \boldsymbol{\theta}_i^{(\ell)} &= \boldsymbol{\theta}_0^{(\ell)} + \pi \tanh(\boldsymbol{\delta}_\theta), \end{aligned} \tag{8}$$

where $\omega_{\max} = 2\pi$ and $\gamma_{\max} = 2$ in normalized time. Amplitude A, latent wavenumber k, angular frequency ω, damping γ, and phase θ are therefore recomputed from the current content at each layer. They are not recomputed within the interval governed by Eq. (6).

4.3. Transition-Local Zero-Time Wave Increment

The transition-local specialization initializes a wave from each state–action pair and does not carry that wave to the next training sample. At layer ℓ,

$$\boldsymbol{\psi}_{i,0}^{(\ell)} = \mathbf{A}_i^{(\ell)} \odot \mathbf{c}_i^{(\ell)} \odot \exp\Big\{\mathrm{i}\left[\mathbf{k}_i^{(\ell)} \odot \mathbf{c}_i^{(\ell)} + \boldsymbol{\theta}_i^{(\ell)}\right]\Big\}. \tag{9}$$

For L stacked layers, each layer advances the wave by $u_i = \widetilde{\Delta t}_i / L$ and reads the change from its zero-time state:

$$\begin{aligned} \boldsymbol{\delta\psi}_i^{(\ell)} &= \left[\mathcal{P}_i^{(\ell)}(u_i) - \mathbf{I}\right] \boldsymbol{\psi}_{i,0}^{(\ell)}, \\ \mathbf{h}_i^{(\ell)} &= \mathbf{h}_i^{(\ell-1)} + \mathbf{W}_r^{(\ell)} \,\mathrm{vec}_{\mathbb{R}}\Big(\boldsymbol{\delta\psi}_i^{(\ell)}\Big), \end{aligned} \tag{10}$$

where $\mathrm{vec}_{\mathbb{R}}(\psi)$ stores both real and imaginary quadratures in a fixed ordering matched to the readout weights. With $\mathbf{h}_i^{(0)} = \mathbf{E}[\mathbf{s}_i; \mathbf{a}_i] + \mathbf{b}_E$, the normalized displacement prediction is

$$\widehat{\mathbf{y}}_i = \mathbf{W}_o\left[\mathrm{LN}(\mathbf{h}_i^{(L)}) - \mathrm{LN}(\mathbf{h}_i^{(0)})\right] - \boldsymbol{\mu}_{\Delta s} \oslash \boldsymbol{\sigma}_{\Delta s}. \tag{11}$$

At $\Delta t_i = 0$, every wave increment is zero. Decoding Eq. (11) with the training target statistics therefore gives exactly zero displacement in raw state units. This specialization evaluates the elapsed-time wave operator

without across-observation carry. The primary sequence and rollout results instead use the persistent model below; transition-local validation diagnostics are distinguished in the appendix.

The evaluated head in Eq. (11) is bias-free and linear, without a separate ReLU/GELU multilayer decoder. This is an architectural choice for the displacement task, not a claim that every application of WSP can omit a task-specific decoder.

4.4. Persistent Injection-Then-Propagation Recurrence

The ordered-window study evaluates a distinct persistent recurrence. From the normalized content, a learned drive and gate form

$$\begin{aligned} \mathbf{d}_i^{(\ell)} &= \mathbf{A}_i^{(\ell)} \odot \left(\mathbf{D}^{(\ell)}\mathbf{c}_i^{(\ell)} + \mathbf{b}_D^{(\ell)}\right), & \mathbf{g}_i^{(\ell)} &= \sigma\left(\mathbf{G}^{(\ell)}\mathbf{c}_i^{(\ell)} + \mathbf{b}_g^{(\ell)}\right), \\ \mathbf{s}_i^{(\ell)} &= \mathbf{d}_i^{(\ell)} \odot \exp\left\{\mathrm{i}\left[\mathbf{k}_i^{(\ell)} \odot \mathbf{c}_i^{(\ell)} + \boldsymbol{\theta}_i^{(\ell)}\right]\right\}. \end{aligned} \tag{12}$$

The current content is injected before physical-time propagation:

$$\begin{aligned} \overline{\boldsymbol{\psi}}_i^{(\ell)} &= \boldsymbol{\psi}_{i-1,+}^{(\ell)} + \mathbf{g}_i^{(\ell)} \odot \mathbf{s}_i^{(\ell)}, \\ \boldsymbol{\psi}_{i,+}^{(\ell)} &= \mathcal{P}_i^{(\ell)}(\widetilde{\Delta t}_i)\overline{\boldsymbol{\psi}}_i^{(\ell)}, \\ \mathbf{h}_i^{(\ell)} &= \mathbf{h}_i^{(\ell-1)} + \mathbf{W}_r^{(\ell)} \operatorname{vec}_{\mathbb{R}}\left(\boldsymbol{\psi}_{i,+}^{(\ell)} - \overline{\boldsymbol{\psi}}_i^{(\ell)}\right). \end{aligned} \tag{13}$$

Here "+" denotes the state after propagation over transition i, not a positive-frequency component. The sequence study uses two layers, each with the full normalized interval. Its reset ablation sets the incoming carry to zero at every transition; its carry model passes $\boldsymbol{\psi}_{i,+}^{(\ell)}$ to the next observation. The persistent model uses the readout in Eq. (11) with its recurrent $\mathbf{h}_i^{(L)}$, preserving zero raw displacement at zero interval. Carry is initialized to zero at the start of each evaluation window; the benchmark does not assume persistence between windows.

4.5. Exact Paired-Real Implementation

Complex arithmetic is not required. For $\psi = u + \mathrm{i}v$, define $\mathbf{z} = [u, v]^\top$ and

$$\mathbf{R}(\phi) = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}. \tag{14}$$

Here $u_i = \widetilde{\Delta}t_i/L$ for the transition-local variant and $u_i = \widetilde{\Delta}t_i$ for the persistent variant. Then one WSP channel is implemented exactly as

$$\mathbf{z}^{(\ell)}_{i,+,m} = \exp\!\big(-\gamma^{(\ell)}_{i,m} u_i\big)\, \mathbf{R}\big(-\omega^{(\ell)}_{i,m} u_i\big)\, \bar{\mathbf{z}}^{(\ell)}_{i,m}. \tag{15}$$

Thus the full WSP uses real tensors but preserves both quadratures. The "cosine-only" ablation discards the sine quadrature and is not the exact real representation of the complex operator.

4.6. Properties and Parallel Scan

For a fixed conditioned generator,

$$\mathcal{P}(0) = \mathbf{I}, \qquad \mathcal{P}(u_2)\mathcal{P}(u_1) = \mathcal{P}(u_1 + u_2). \tag{16}$$

When the conditioner changes between observations, the second identity does not imply one global semigroup; it applies only while the generator is held fixed. For nonnegative damping and $u \geq 0$,

$$\|\mathcal{P}(u)\|_2 \leq 1. \tag{17}$$

Equality is possible for an undamped channel. This is a bound on the homogeneous flow, not on arbitrary driven input. For the affine recurrence $\boldsymbol{\psi}_i = \mathbf{P}_i \boldsymbol{\psi}_{i-1} + \mathbf{b}_i$ with $\|\mathbf{P}_i\|_2 \leq 1$, a simple driven bound is $\|\boldsymbol{\psi}_n\|_2 \leq \|\boldsymbol{\psi}_0\|_2 + \sum_{i=1}^{n} \|\mathbf{b}_i\|_2$.

With conditioner outputs held fixed, the direct derivatives of one evaluated operator entry are

$$\frac{\partial [\mathcal{P}_i]_{mm}}{\partial \omega_{i,m}} = -\mathrm{i} u_i [\mathcal{P}_i]_{mm}, \qquad \frac{\partial [\mathcal{P}_i]_{mm}}{\partial \gamma_{i,m}} = -u_i [\mathcal{P}_i]_{mm}. \tag{18}$$

Wavenumber k receives gradients through Eq. (9) or Eq. (12), not through the evaluated physical-time operator.

For the persistent injection-before-propagation rule, define $\mathbf{b}_i := \mathbf{P}_i(\mathbf{g}_i \odot \mathbf{s}_i)$. Two affine steps compose as

$$(\mathbf{P}_2, \mathbf{b}_2) \circ (\mathbf{P}_1, \mathbf{b}_1) = (\mathbf{P}_2\mathbf{P}_1, \mathbf{P}_2\mathbf{b}_1 + \mathbf{b}_2). \tag{19}$$

This operation is associative, so a layer can in principle use a parallel prefix scan. The reported persistent implementation is serial; the scan is a mathematical implementation path and is not used to support the measured throughput claims.

All reported WSP models are trained end-to-end with the normalized displacement loss and ordinary backpropagation. No wave-equation residual, dispersion penalty, energy penalty, or Laplacian parameter update is used. Transformer fusion and phase-aware attention are possible extensions, but they are outside the evaluated architecture and are therefore not included in the claimed method.

5. Evaluation

5.1. Study Design and Prediction Task

The primary evaluation compares the persistent WSP recurrence with five temporal sequence models on four preselected datasets. It asks whether WSP improves prediction under a shared parameter budget, uses informative elapsed-time inputs, maintains accuracy during open-loop replay, and offers a favorable inference-memory–throughput trade-off. Every model predicts the same normalized next-state displacement from the current state, available action, causal history, and upcoming elapsed interval. This is supervised sequence prediction, not reinforcement-learning return or closed-loop control evaluation.

Three forms of evidence are kept distinct. The primary benchmark uses ten final training seeds after validation-only model selection. An independent five-seed component study changes individual WSP computations under its own fixed configuration. Additional development diagnostics compare sequence and transition-local models using validation data only (Appendix Tables F.11 and F.12). Validation rankings, component effects, and final test results are not pooled across these protocols.

5.2. Datasets and Sequence Construction

The *damped oscillator* is a controlled continuous-time dynamics task. *HalfCheetah-v5* supplies articulated-body dynamics from MuJoCo [22]. Actions are held over a known number of physics steps, so $\Delta t_i = n_i h$ for native integration step h. Counterfactual branches share an anchor state and action but vary the interval; only the behavior branch advances an episode and enters an ordered sequence. Training, validation, and test episodes use separate seeds.

The *PhysioNet2012* adaptation uses irregular physiological records from Set A of the PhysioNet/Computing in Cardiology Challenge 2012 [23, 24].

The adapter forms a multivariate state at the union of selected observation times. It carries previous measurements forward and initializes not-yet-observed variables using medians fitted on training patients. The ten channels cover heart rate, systolic/diastolic/mean blood pressure, respiratory rate, temperature, oxygen saturation, urine, weight, and Glasgow Coma Scale. Time stamps are converted to seconds; the task has a zero-valued action channel. Patient-disjoint partitions prevent a patient's sequence from crossing splits. All models receive the same prepared values without an observation mask. The objective is dense state-displacement prediction, not the Challenge's mortality outcome or an observation-mask-aware imputation score.

The Δt-*irrelevant control* uses a nonlinear discrete state–action transition whose target does not depend on the supplied time interval. Independently generated intervals accompany the transitions. This tests sensitivity to nuisance timing; it does not require a time-aware architecture to have poor predictive accuracy.

The primary benchmark uses contiguous windows of 32 behavior transitions. Training stride is 16 and evaluation stride is 32. Windows do not cross episode boundaries, and shorter tails are excluded. Each model resets to its prescribed initial state at each window; WSP carry starts from zero, while f-CRU also initializes its covariance. Thus the experiment tests within-window sequence learning, not indefinite state persistence.

State, action, and displacement normalization uses training statistics only. For state coordinate j, the prediction target is

$$y_{i,j} = \frac{\Delta s_{i,j} - \mu_{\Delta s,j}}{\sigma_{\Delta s,j}}, \qquad \Delta \mathbf{s}_i = \mathbf{s}_{i+1} - \mathbf{s}_i. \tag{20}$$

The propagation time scale T is derived from the training intervals.

5.3. Comparators, Optimization, and Statistical Analysis

The primary WSP model uses width 128, two persistent layers, and a rank-16 conditioner. Comparator widths are selected to match its trainable parameter count within a 5% tolerance; the achieved discrepancy is at most 0.915%. Parameter counts are approximately 158,000–165,000, depending on dataset and model (Appendix Table A.1). Parameter matching controls capacity approximately, not computational work or optimization difficulty.

The comparators are *f-CRU*, *LRU*, *Phased LSTM*, *UnICORNN*, and a *time-conditioned, zero-time-constrained Transformer*. The f-CRU adaptation

carries a Gaussian latent mean and structured covariance through observation assimilation and closed-form interval propagation [15]. LRU retains its stable complex diagonal discrete recurrence, with elapsed time supplied as an input feature [8]. Phased LSTM gates its hidden and cell updates using an oscillatory clock [14]; the clock restarts with each window. UnICORNN retains its oscillator recurrence and uses physical time as an input feature rather than replacing its numerical integration step by Δt [16].

The Transformer uses causal attention, learned positional embeddings, and time features. Its local readout subtracts the same head evaluated at zero current interval, enforcing zero raw displacement at $\Delta t = 0$. Its context incorporates previous intervals, not future observations. Consequently, this comparison does not attribute a WSP advantage merely to the absence of a zero-time boundary in the Transformer.

These models are independent, task-adapted implementations of the cited recurrences, not reproductions of their native tasks or optimized kernels. For example, f-CRU uses the common displacement MSE rather than its original probabilistic output objective; UnICORNN uses eager PyTorch autograd rather than its authors' fused reversible implementation. Native architecture hyperparameters and model-specific optimizer parameter groups are not exhaustively tuned.

Each model receives the same learning-rate search $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ with weight decay 10^{-4}, development seeds $\{1, 2\}$, at most 50 epochs, and early-stopping patience 10. The setting with the lowest mean validation NRMSE is locked before final training. Final runs use seeds 101–110, at most 100 epochs, and patience 20. All runs use AdamW, cosine learning-rate decay, gradient clipping at 1, batch size 256, and the same normalized-displacement MSE. Selected settings are reported in Appendix Table A.2.

Training is distributed as independent single-GPU jobs across eight NVIDIA B200 GPUs, using Python 3.11.11, PyTorch 2.9.1 with CUDA 13.0, and unfused AdamW. TF32 is disabled. BF16 denotes a mixed-precision policy, not uniform arithmetic: sensitive recurrences retain FP32 and f-CRU runs entirely in FP32.

For N evaluated transitions with D target coordinates,

$$\mathrm{NRMSE}_{\Delta s} = \sqrt{\frac{1}{ND} \sum_{i=1}^{N} \sum_{j=1}^{D} (\widehat{y}_{i,j} - y_{i,j})^2}. \tag{21}$$

We first aggregate prediction elements within a seed, then report the mean

and sample standard deviation across ten seeds. NRMSE is dimensionless error in training-standardized displacement space, not an accuracy percentage. A value above one is not by itself evidence of being worse than a persistence baseline, particularly under a shifted test-time distribution.

Model comparisons pair the same seed across architectures. We report 95% nonparametric bootstrap intervals from 20,000 paired resamples and two-sided sign-flip tests. Holm adjustment covers the 20 primary dataset–baseline comparisons. The seed-level uncertainty describes training variability on fixed data partitions, not variation across new patient cohorts or independently generated datasets.

5.4. Primary Predictive Accuracy

Figure 1 shows all six models without combining their four dataset scales. WSP has the lowest mean test NRMSE on HalfCheetah, the oscillator, and the time-irrelevant control; LRU is lowest on PhysioNet. Appendix Tables A.1 and B.3 provide the exact means, standard deviations, paired differences, confidence intervals, and adjusted tests.

HalfCheetah gives the clearest positive result. WSP reaches 0.71079 ± 0.01100, compared with 0.97622 ± 0.04279 for LRU and 1.09851 ± 0.06928 for Transformer, reductions of 27.2% and 35.3%, respectively. WSP outperforms each comparator on all ten paired seeds, with Holm-adjusted $p = 0.0391$ for each comparison.

On the oscillator, WSP reaches 1.83733 ± 0.01738 versus 1.86019 ± 0.14590 for Transformer. The 1.23% difference in means is not statistically resolved: WSP wins five of ten seeds and the 95% paired interval for Transformer-minus-WSP is $[-0.0633, 0.1076]$. WSP nevertheless has lower error than the four recurrent comparator adaptations under the reported corrected tests.

On PhysioNet, WSP reaches 0.90859 ± 0.00166, close to f-CRU (0.90909 ± 0.00187) and Transformer (0.90913 ± 0.00163); neither paired difference is statistically resolved. LRU reaches 0.90549 ± 0.00175, a small but consistent advantage: its NRMSE is 0.34% lower than WSP's and it wins all ten seeds. This is an important boundary on the empirical claim, not a general clinical superiority result.

On the time-irrelevant control, WSP reaches 0.01687 ± 0.00060 and substantially outperforms the tested comparators. Thus a physical-time parameterization does not force poor performance when the supplied timing is irrelevant. The appropriate complementary question is whether the trained model unnecessarily depends on that timing.

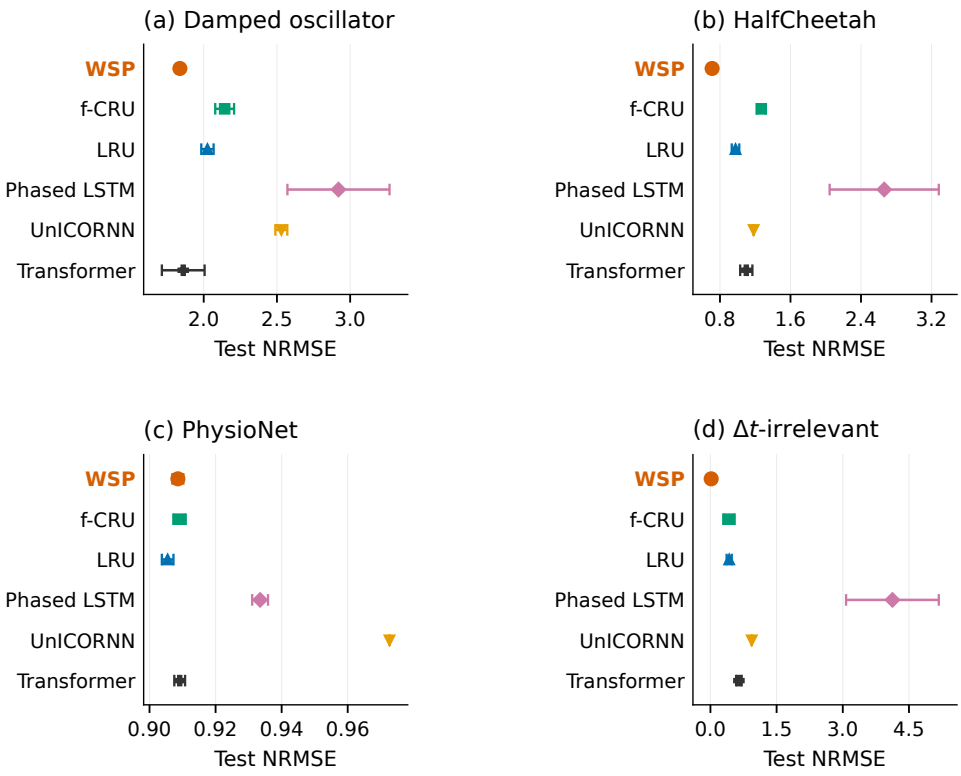


Figure 1: Primary test displacement NRMSE. Points and bars show the mean and sample standard deviation over ten final training seeds. Lower is better. Each panel has its own linear horizontal scale; the PhysioNet scale is expanded to show small differences. All models use the same task-specific parameter budget. Error-bar overlap is not used as a significance test; paired results are reported in the appendix.

5.5. Learning-History Audit

Because all ten HalfCheetah LRU and Transformer runs select epochs 1–2, we inspect their recorded optimization histories alongside WSP (Figure 2). Across the actual stopping epochs, LRU's mean online training MSE falls from 0.5149 to 0.0454, and Transformer's from 0.6202 to 0.0293. Their mean validation NRMSE nevertheless rises from the selected minima of 0.5365 and 0.5761 to 0.8527 and 0.8686, respectively. This train–validation divergence is consistent with early overfitting or a training–validation mismatch, not a flat training-loss trajectory. It does not identify the cause or establish optimal tuning.

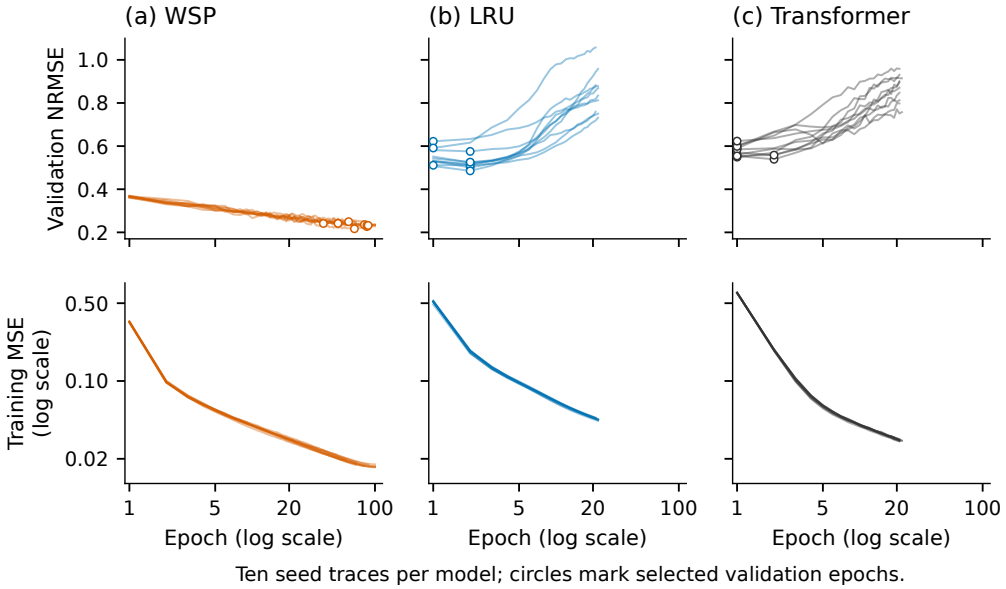


Figure 2: HalfCheetah learning histories for WSP, LRU, and Transformer. Each thin line is one of ten final seeds and ends at its actual stopping epoch; circles identify the validation-selected checkpoint. All epoch axes are logarithmic. Online training MSE is shown separately from post-epoch validation NRMSE and is not directly equated with it. No trace is extended beyond stopping or averaged over a changing set of surviving runs.

WSP selects epochs 38–88, with mean validation NRMSE 0.2353 at selection and 0.2373 at stopping. All reported test scores use the selected checkpoint, not the final, potentially overfit iterate. The audit also exposes distinct behaviors among the other baselines: f-CRU has lower selected validation NRMSE than WSP (0.2279) despite its worse time-shifted test result, whereas UnICORNN's online training MSE decreases by only 0.7% on average. Consequently, validation generalization, test-time extrapolation, and weak optimization progress should not be treated as the same limitation. Appendix Tables H.16 and H.17 give all six models' histories in summary. The three-rate search and early stopping constrain the comparison but do not establish equally optimized model families; these curves motivate model-specific tuning without changing the already reported test selection.

5.6. Temporal Sensitivity and Generalization

For a fixed trained model and unchanged targets, we replace elapsed-time inputs by shuffled, constant, zero, halved, doubled, or noisy intervals. The primary timing-reliance measure is

$$D_{\text{shuffle}} = \text{NRMSE}_{\Delta s}(\text{shuffled time}) - \text{NRMSE}_{\Delta s}(\text{correct time}). \tag{22}$$

Figure 3 reports WSP's paired bootstrap intervals. Shuffling increases mean NRMSE by 0.10296 on the oscillator, 0.10123 on HalfCheetah, and 0.00589 on PhysioNet, corresponding to 5.6%, 14.2%, and 0.65% of their correct-time mean errors. The time-irrelevant change is -0.000077, with interval $[-0.000227, 0.000077]$, consistent with negligible shuffled-time dependence.

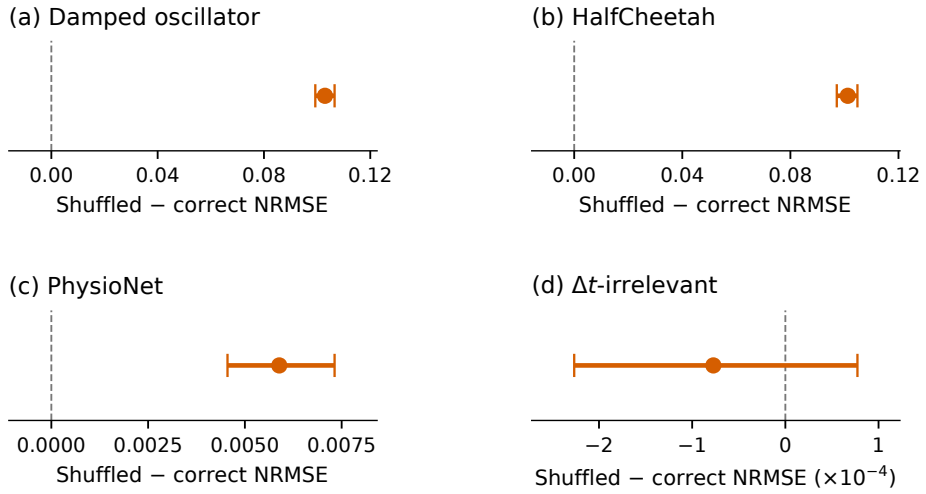


Figure 3: WSP sensitivity to shuffled elapsed-time inputs. Points show paired mean changes in test NRMSE and bars show 95% bootstrap intervals across ten seeds. Positive changes indicate deterioration. Panel scales differ to retain the small PhysioNet and negative-control effects; zero is marked explicitly. This intervention tests input reliance, not the isolated causal contribution of a particular WSP component.

The negative control is not perfectly invariant to every manipulation: constant timing improves its NRMSE from 0.01687 to 0.01393. Moreover, the zero-time intervention structurally forces WSP's raw displacement to zero while retaining the original nonzero target. Its large error increase is therefore not clean evidence of learned time dependence. Shuffled-time results

provide the more informative comparison. The full intervention values are in Appendix Tables C.5 and C.6.

Figure 4 separates error by the ratio $\Delta t/T$. An interval is *seen* if it approximately matches a training interval; an unseen interval inside the training range is interpolation, and one above the largest training interval is long-gap extrapolation. These labels concern time values, not reuse of training episodes or patients.

Long gaps constitute 43,908 of 51,200 oscillator test steps and 82,150 of 96,000 HalfCheetah test steps, approximately 86% in each case. WSP's interpolation/extrapolation NRMSE is 0.11339/1.98350 for the oscillator and 0.27635/0.75995 for HalfCheetah. The corresponding Transformer values are 0.23638/2.00634 and 0.51220/1.16860. This test-time shift helps explain the large validation–test gap and shows that low interpolation error does not guarantee accurate long-gap prediction.

The relative-error panels make the scope of the advantage more precise. On HalfCheetah, WSP has lower NRMSE than both LRU and Transformer on all ten seeds at each of the seven tested intervals. Its reductions in mean error range from 17.5–47.9% against LRU and 26.2–50.9% against Transformer. The advantage is not monotonic in elapsed time. On the oscillator, the Transformer-relative reduction is 52.0% at $\Delta t/T = 0.375$ but becomes -6.8% at $\Delta t/T = 5$: WSP is worse at that largest interval, with lower error in only two of ten seeds. Thus informative timing and a favorable overall mean do not imply increasingly strong extrapolation at every gap. Appendix Table J.20 reports all interval-specific contrasts; their pointwise intervals are descriptive follow-up analyses, not additions to the 20 corrected primary tests.

PhysioNet's evaluated time categories contain 27,452 seen-gap steps, 35 interpolation steps, and one long-gap step per seed. Its aggregate result therefore assesses prediction mainly at previously represented gap values; it does not establish robust clinical long-gap extrapolation. Condition-specific means and counts are reported in Appendix Table C.4.

5.7. Prediction Across Window Prefixes

To examine how errors vary within the evaluated windows, we score the first $L \in \{1, 4, 8, 16, 32\}$ transitions of each original test window using the same selected checkpoint and per-window initialization as in the primary evaluation. This reuses the recorded prefix evaluations; it does not retrain a model for each length. Crucially, all L prefix targets are scored. Changing L

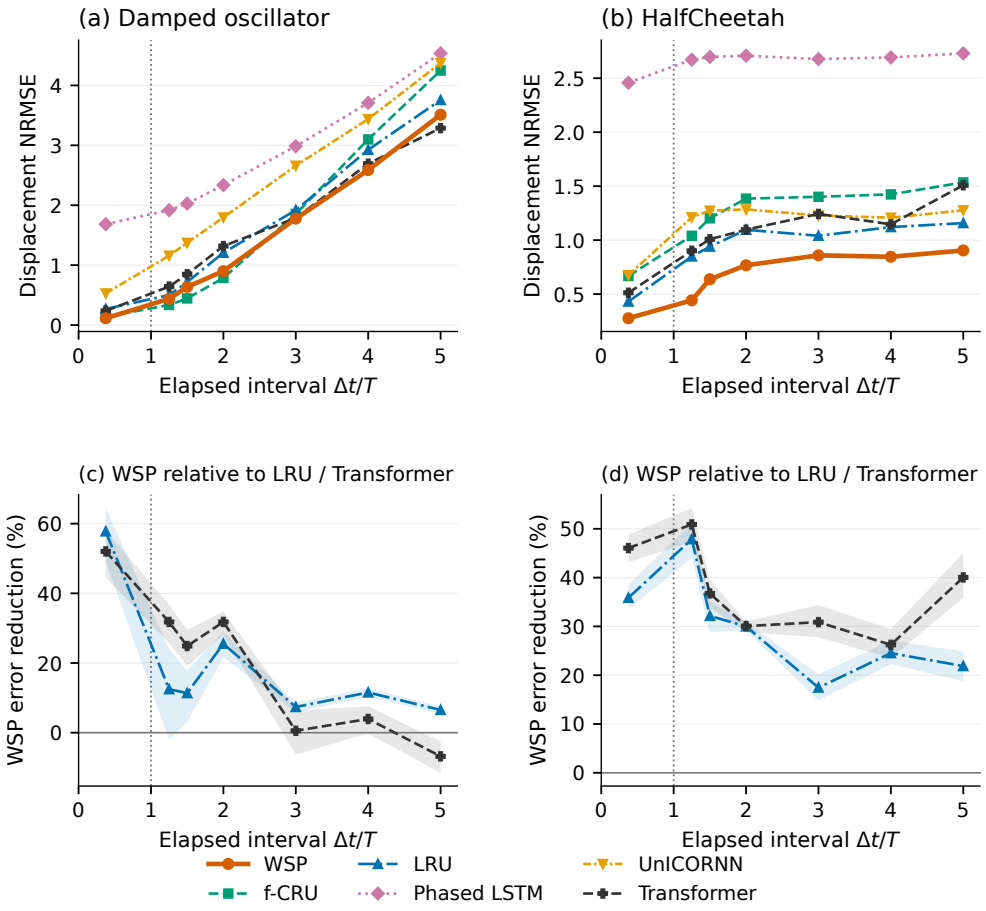


Figure 4: Elapsed-time-resolved prediction on the physical-dynamics tasks. Top: mean displacement NRMSE over ten seeds for all six models. Bottom: WSP's relative error reduction against LRU and Transformer, $100(\bar{e}_b - \bar{e}_{\mathrm{WSP}})/\bar{e}_b$; positive values favor WSP. Shading gives pointwise 95% paired bootstrap intervals, without multiplicity adjustment, for this exploratory diagnostic. The dotted vertical line marks the largest training interval, $\Delta t/T = 1$. Individual-gap, aggregate extrapolation, and overall test scores need not have the same ranking.

therefore changes both available history and the evaluated target population, unlike a fixed-target history-truncation ablation.

WSP's mean NRMSE decreases from $L = 1$ to $L = 32$ in all four datasets (Figure 5). On HalfCheetah it changes from 0.77416 to 0.71079 and is lowest among the six models at every scored prefix. On the oscillator, however, WSP

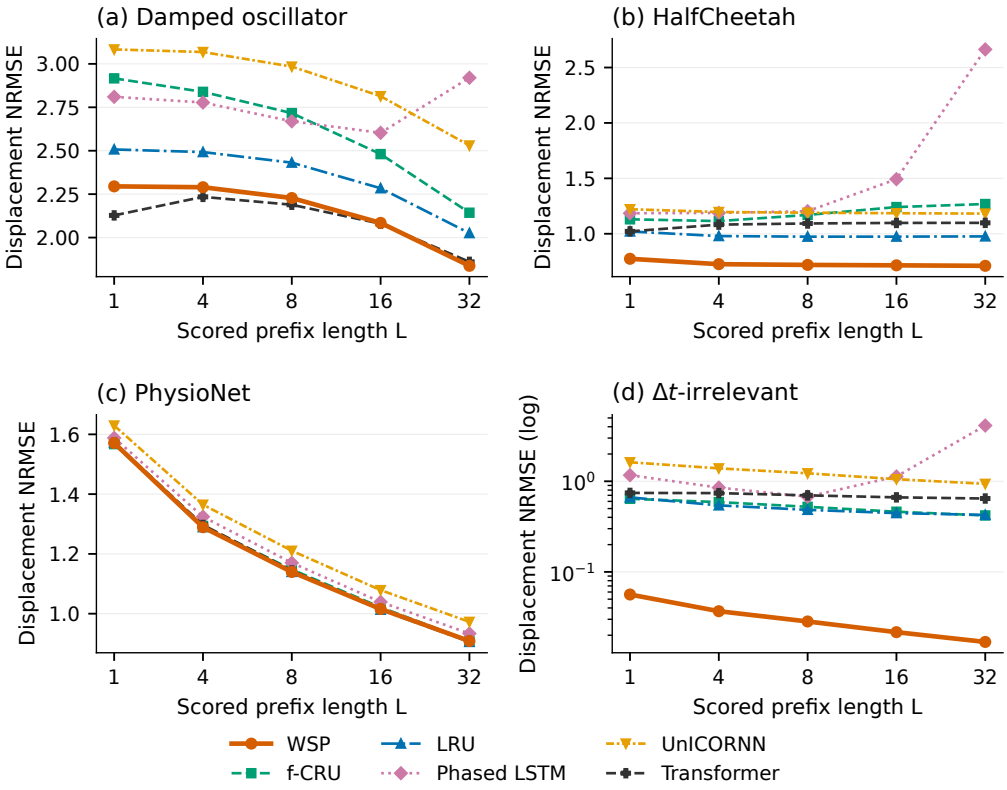


Figure 5: Mean test displacement NRMSE across scored window prefixes, using ten final seeds per model. Lengths are discrete evaluation settings; the time-irrelevant panel uses a logarithmic vertical scale. All models score identical targets at a given L, but target sets differ between lengths. These curves describe within-window prediction, not the isolated causal effect of additional memory.

is worse than Transformer at $L = 1$ (2.29477 versus 2.12755), while their full-window means are close. On PhysioNet, all six models show substantial reductions across prefixes; that shared pattern cannot be credited uniquely to WSP memory. The prefix analysis therefore supports robust HalfCheetah prediction within the evaluated windows, while the separate carry-versus-reset study in Section 5.9 is the relevant controlled test of state persistence. Appendix Table J.19 gives the prefix means, seed standard deviations, and scored-element counts.

5.8. Open-Loop Autoregressive Replay

We feed each predicted state back to the model while replaying recorded actions and elapsed intervals, starting from an observed state. The horizons are 4, 8, 16, and 32 transitions. The rollout score averages errors over all steps up to horizon H:

$$\mathrm{NRMSE}_s(H) = \sqrt{\frac{1}{N_w H D} \sum_{w=1}^{N_w} \sum_{h=1}^{H} \sum_{j=1}^{D} \left(\frac{\widehat{s}_{w,h,j} - s_{w,h,j}}{\sigma_{s,j}} \right)^2}, \tag{23}$$

where $\sigma_{s,j}$ is the training state standard deviation. It is not an endpoint-only metric and is not numerically interchangeable with displacement-normalized Eq. (21).

Figure 6 shows that the one-step ranking does not universally carry over to replay. At 32 steps, HalfCheetah WSP reaches 0.70487 versus 0.83243 for LRU and 0.99949 for Transformer. On the oscillator, f-CRU is better than WSP (0.75705 versus 0.83528). On PhysioNet, LRU is best among these models (0.91793), while WSP reaches 0.93329. WSP has the lowest mean replay error on the time-irrelevant control. Exact values and seed standard deviations are in Appendix Table D.7. Neither these finite windows nor the homogeneous operator bound proves long-horizon stability of the complete driven predictor.

5.9. Independent Component Diagnostics

A separate five-seed study compares persistent WSP with controlled computational ablations on Reacher, HalfCheetah, the oscillator, and Hopper. The first three tasks use 32-transition windows; Hopper uses four-transition windows. These settings and comparators differ from the primary benchmark, so absolute NRMSE values are not pooled across studies. Appendix G gives the exact configurations and tables.

Carry lowers mean error relative to resetting at every transition by 7.4% on Reacher, 39.9% on HalfCheetah, 2.3% on the oscillator, and 2.5% on four-step Hopper. Exploratory paired intervals exclude zero in Reacher, HalfCheetah, and Hopper, but not in the oscillator. Damping is beneficial in all four tested settings, while frequency, quadrature, conditioning, and zero-time ablations have mixed effects. For example, removing frequency improves the oscillator and short-Hopper means. These controls support useful contributions from persistence and damping in the tested configurations, not universal

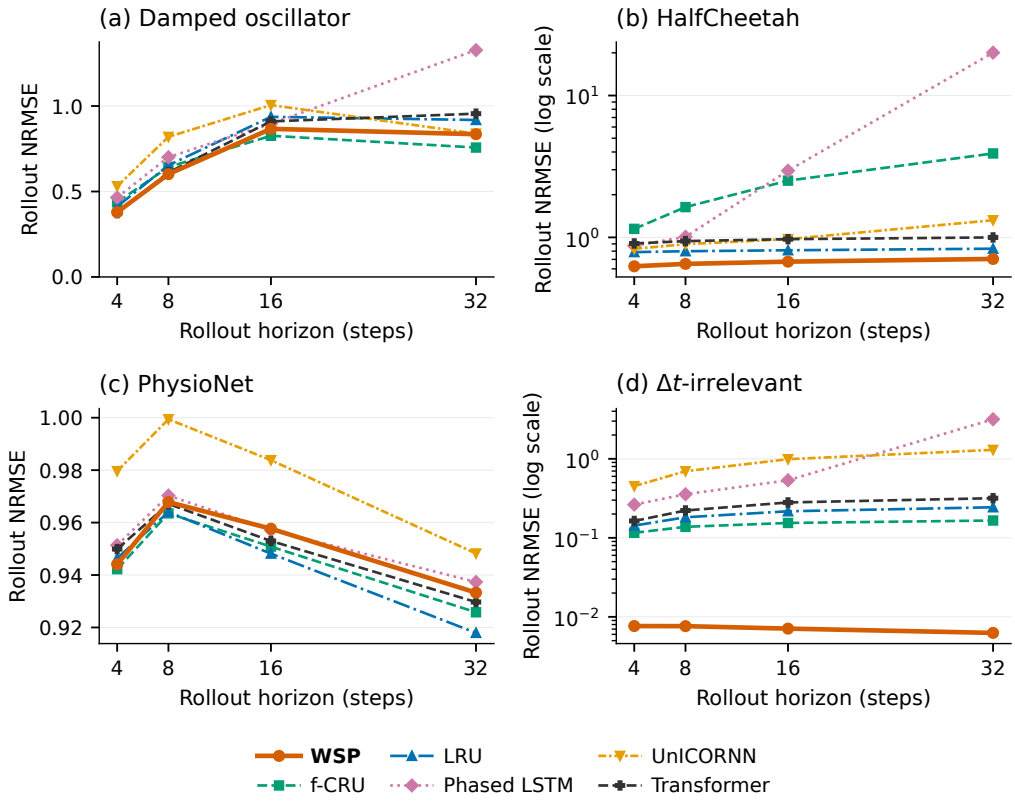


Figure 6: Open-loop state-normalized replay error by horizon, with all six models. Lines summarize ten seeds. Logarithmic vertical scales are used where errors span orders of magnitude and are identified on the axes. The policy is not recomputed from predicted states; these are not closed-loop control returns.

necessity of every wave parameter. They do not isolate component effects on PhysioNet or the time-irrelevant control.

5.10. Comparison with Diagonal-State-Space Controls

The development study provides a closer architectural comparison with global and content-conditioned diagonal SSMs. Both controls use an affine projection into paired real quadratures, damped rotation, and the same injection–propagation–difference and zero-time output shell as WSP. The global control learns shared frequencies and damping rates; the conditioned control adapts them to content. WSP instead uses its content-dependent wave injection with amplitude, phase, and a gated affine drive. Widths are

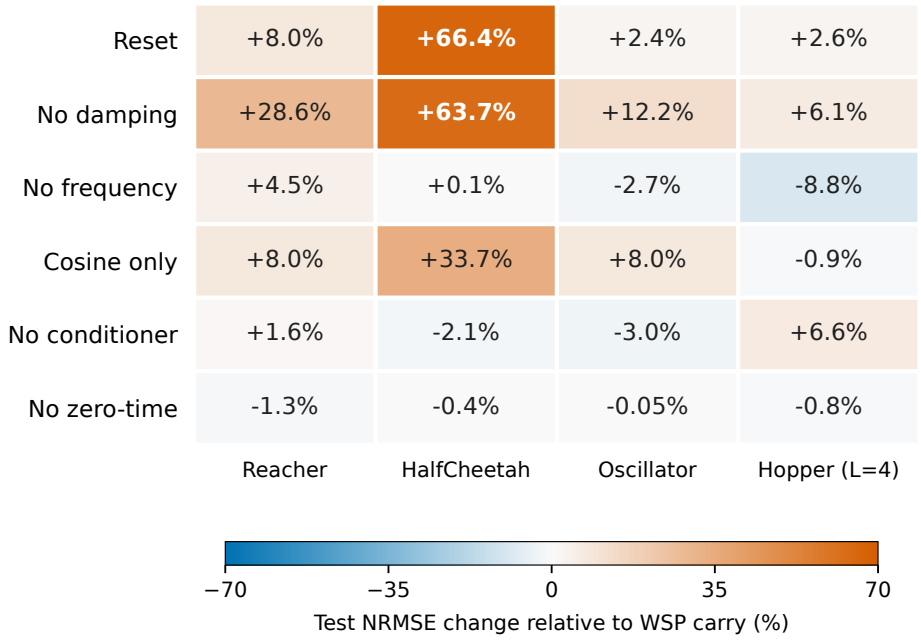


Figure 7: Independent five-seed WSP component diagnostics. Cells show $100(e_{\text{ablation}} - e_{\text{carry}})/e_{\text{carry}}$ using this study's mean test NRMSE. Reacher, HalfCheetah, and the oscillator use length 32, whereas Hopper uses length 4. Positive changes indicate deterioration. "No zero-time" removes the displacement constraint; it is distinct from replacing the elapsed-time input by zero. These values are not pooled with the ten-seed primary benchmark.

adjusted to match capacity (128 for WSP, 140 for global SSM, and 134 for conditioned SSM), with parameter discrepancies below 0.64%. This is a matched-capacity architectural control, not an isolated one-parameter ablation or a comparison with every nonlinear complex injection.

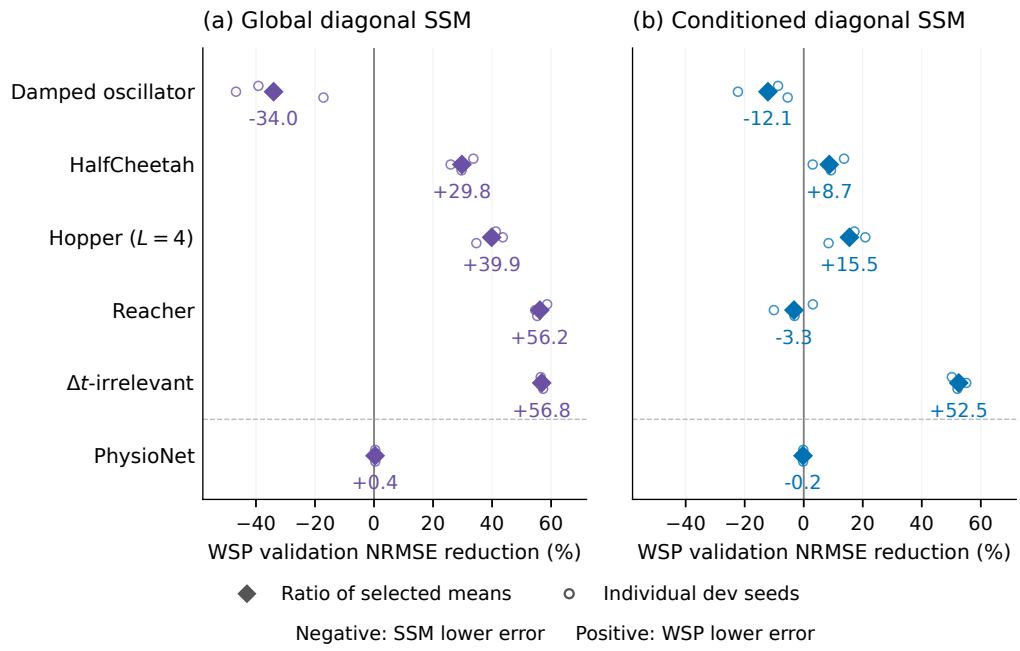


Figure 8: Development-only comparison with diagonal-SSM controls. Diamonds show $100(\bar{e}_{\mathrm{SSM}} - \bar{e}_{\mathrm{WSP}})/\bar{e}_{\mathrm{SSM}}$ from the selected validation means; open points show the three individual development-seed contrasts. Positive values favor WSP. These are not test-set confidence intervals: both configuration and epoch were selected on the same validation data. Hopper uses $L = 4$ and all other tasks $L = 32$. The separated PhysioNet row uses its independent development preparation.

WSP reduces selected mean validation error relative to the conditioned SSM by 8.7% on HalfCheetah, 15.5% on short-window Hopper, and 52.5% on the time-irrelevant control. Conversely, the conditioned SSM is better on the oscillator, Reacher, and PhysioNet; the global SSM is also better on the oscillator (Figure 8). These mixed contrasts show that useful paired-real dynamics do not require WSP's specific wave injection, and that its advantage is task-dependent. They do not establish an independent test benefit of the injection alone. Appendix Table I.18 gives the exact relative contrasts; Table F.11 gives their absolute validation errors and selected settings. Selection optimism and the three-seed development protocol preclude pooling these values with the ten-seed primary test benchmark.

5.11. Accuracy-Matched Inference Efficiency

We benchmark the 240 trained primary checkpoints on an idle, exclusive NVIDIA B200 at sequence length 32 and batch sizes 1, 16, and 256. Each workload has ten warm-up passes and 30 synchronized measured passes, giving 720 checkpoint–workload combinations and 21,600 latency samples. Inference uses deterministic shape-matched synthetic inputs with the checkpoint's normalization and time scale. The measurement is forward-pass runtime, not end-to-end data loading, host transfer, or application latency. For a seed-specific mean latency $\bar{t}$, throughput is $32B/\bar{t}$ transitions per second; displayed values average seed-specific throughputs.

Peak allocated memory is PyTorch's maximum allocated device memory with the model, inputs, buffers, and temporary activations resident. Values use MiB (2^{20} bytes), exclude reserved-cache capacity and CUDA-context overhead, and are distinct from both incremental activation memory and training peaks. Runtime policy and implementations are the same declared ones used in the benchmark; f-CRU retains its FP32 covariance computation and UnICORNN does not use a fused reversible kernel.

A workload is validation-accuracy matched when both models have the same ten-seed set and

$$100\,\frac{|\bar{e}^{val}_{\text{candidate}} - \bar{e}^{val}_{\text{WSP}}|}{|\bar{e}^{val}_{\text{WSP}}|} \leq 1, \tag{24}$$

where e is validation NRMSE. Test errors are reported after selection, not used to form the match. This operational tolerance is neither exact per-seed equality nor a statistical equivalence test.

The matched comparisons occur on PhysioNet against f-CRU, LRU, and Transformer. Repeating these three comparisons at three batch sizes gives nine matched workloads, not nine independent accuracy comparisons. Their relative validation gaps are 0.322%, 0.972%, and 0.144%, respectively. The other datasets are not labeled accuracy matched.

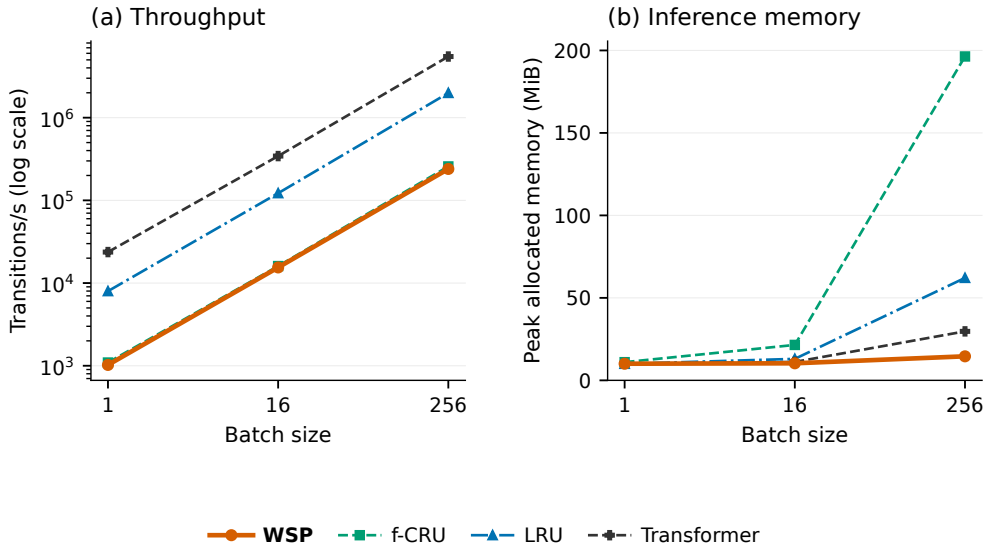


Figure 9: Validation-accuracy-matched PhysioNet inference at length 32. Throughput and total peak allocated memory are shown separately across batch sizes. Points summarize ten trained checkpoints per model. Higher throughput and lower memory are favorable, but WSP trades speed for lower memory at larger batches. The matching tolerance is defined in Eq. (24); it does not assert exact accuracy equality.

At batch 256, WSP processes 239,842 transitions/s with a 14.57 MiB peak, compared with 5,488,554 transitions/s and 29.69 MiB for Transformer. Thus WSP uses 50.9% less allocated inference memory but is 22.9 times slower. Against LRU it uses 76.5% less memory but is 8.3 times slower. Against f-CRU it uses 92.6% less memory with 6.7% lower throughput. The Transformer-relative memory reduction is batch-dependent: 0.24% at batch 1, 7.45% at batch 16, and 50.94% at batch 256. Detailed operating points are in Appendix Tables E.8 and E.9.

Training and inference costs must also be separated. PhysioNet WSP's reported training peak is approximately 249 MiB versus 93 MiB for Transformer (Appendix Table E.10). The evidence therefore supports a batched

inference-memory benefit, not general memory savings throughout training and deployment. Lower allocation alone does not establish lower monetary cost, energy use, or proportional gains from packing additional replicas onto a GPU.

5.12. Implications for Physical-Time Sequence Learning

Taken together, the primary results support the viability of learning sequence transitions through content-conditioned wave states and explicit physical-time propagation. HalfCheetah provides the clearest evidence: WSP has lower NRMSE than LRU and Transformer at all seven tested intervals, lower mean error at every scored window prefix, and lower mean 32-step recorded-action replay error. Shuffled-time interventions show that the trained model uses informative elapsed intervals, while the time-irrelevant control shows negligible sensitivity to the same intervention. These findings establish a useful empirical role for the architecture without requiring it to rank first on every task.

The architectural contribution is a trainable connection between observed content and elapsed-time evolution. Content-conditioned amplitude and phase construct the injected wave state; learned frequency and damping then determine its evolution over the measured interval. Persistent state connects successive observations, and the zero-time readout maps that evolution to a displacement prediction. This makes WSP a natural candidate for sequence-prediction applications in which the duration between updates has predictive meaning, rather than merely identifying observation order. The benchmark demonstrates this possibility in selected settings; it does not validate every application that provides timestamps.

The supporting analyses help delimit this interpretation. Independent component diagnostics identify benefits from persistence and damping in their tested settings, while the development-only comparisons place WSP alongside competitive alternatives within the continuous-time SSM family. These protocols are not pooled with the primary test benchmark. Learning histories also show that the common training budget produces different optimization and generalization behavior across models. Consequently, the observed rankings characterize the declared implementations and search, rather than an exhaustive comparison of their representational capacity.

For an intended application, predictive error and deployment requirements must be considered jointly. The accuracy-matched PhysioNet measurements show a batched inference-memory benefit, but the current serial

WSP implementation has lower throughput than the compared Transformer and LRU. The associative composition in Eq. (19) motivates numerically validated parallel execution; it is not an existing speedup. Together with the task- and horizon-dependent accuracy rankings, these results support WSP as a viable modeling option whose suitability depends on the target dynamics, sequence length, batch size, numerical backend, and acceptable memory and latency costs.

6. Conclusion and Future Work

WSP provides a wave-parameterized approach to neural sequence prediction with measured continuous physical time. It combines content-conditioned amplitude and phase, closed-form damped rotation, persistent injection-then-propagation, and a zero-time displacement readout. Both quadratures are retained through an exact paired-real implementation, and the complete predictor is trained end-to-end. Building on the continuous-time SSM family, this design makes the role of elapsed time explicit while retaining a learned nonlinear representation of observed content.

The parameter-matched ten-seed benchmark supports the practical promise of this approach. WSP achieves substantial HalfCheetah accuracy gains over the tested temporal baselines and uses informative elapsed intervals, with task-dependent rankings in the other evaluated settings. Independent component diagnostics identify useful contributions from persistence and damping. Accuracy-matched PhysioNet measurements also demonstrate a batched inference-memory benefit, accompanied by lower throughput in the current serial implementation. These results describe a useful modeling and deployment trade-off rather than a uniformly dominant architecture.

The four-dataset benchmark samples physical-dynamics and physiological prediction tasks together with a time-irrelevant control; it does not span the full range of continuous-time applications. Explicit timing alone therefore does not guarantee that WSP is the preferred model. Future work should examine numerically validated parallel execution, longer sequences, model-specific tuning, observation-mask-aware real-data tasks, and closed-loop control. These extensions require application-specific accuracy and efficiency measurements.

The central conclusion is that WSP is a viable approach to sequence learning in applications where measured elapsed time informs the underlying

dynamics. Its value lies in providing an explicit, wave-parameterized model of temporal evolution, with suitability determined by the accuracy, memory, and latency requirements of the intended application.

Appendix A. Detailed primary sequence results

All primary comparisons use the same ten final random seeds (101–110) per model and dataset. Tables report sample standard deviations across seeds, not standard errors. Values are rounded to the displayed precision; a displayed zero SD can denote a small positive SD. Lower NRMSE is better. Transformer denotes the time-conditioned, zero-time-consistent sequence baseline. The four named recurrent competitors are common-task adaptations; their results are not native-benchmark reproductions.

Table A.1: Primary sequence benchmark: test and validation NRMSE (mean ± SD), with trainable parameter counts. Validation values here are measured on the ten final-seed fits; they are distinct from the two-seed scores used to choose the learning rate.

Model	Test NRMSE	Validation NRMSE	Parameters
Damped oscillator			
WSP	1.8373 ± 0.0174	0.1771 ± 0.0113	158,976
f-CRU	2.1431 ± 0.0643	0.0382 ± 0.0083	157,922
LRU	2.0255 ± 0.0421	0.4072 ± 0.1435	159,602
Phased LSTM	2.9202 ± 0.3486	0.4397 ± 0.0372	157,922
UnICORNN	2.5301 ± 0.0408	0.7235 ± 0.0226	158,694
Transformer	1.8602 ± 0.1459	0.3634 ± 0.1477	158,014
HalfCheetah			
WSP	0.7108 ± 0.0110	0.2353 ± 0.0093	163,456
f-CRU	1.2684 ± 0.0471	0.2279 ± 0.0098	162,907
LRU	0.9762 ± 0.0428	0.5365 ± 0.0443	163,607
Phased LSTM	2.6634 ± 0.6193	0.8347 ± 0.0213	162,927
UnICORNN	1.1821 ± 0.0003	0.9723 ± 0.0002	163,258
Transformer	1.0985 ± 0.0693	0.5761 ± 0.0277	164,635

Continued on next page

Table A.1 (continued)

Model	Test NRMSE	Validation NRMSE	Parameters
PhysioNet			
WSP	0.9086 ± 0.0017	0.9179 ± 0.0005	160,896
f-CRU	0.9091 ± 0.0019	0.9149 ± 0.0009	160,060
LRU	0.9055 ± 0.0018	0.9090 ± 0.0003	161,320
Phased LSTM	0.9335 ± 0.0024	0.9476 ± 0.0025	161,962
UnICORNN	0.9726 ± 0.0000	0.9851 ± 0.0000	161,290
Transformer	0.9091 ± 0.0016	0.9166 ± 0.0008	159,424
Δt-*irrelevant*			
WSP	0.01687 ± 0.00060	0.02019 ± 0.00099	160,128
f-CRU	0.41994 ± 0.12855	0.04072 ± 0.00092	159,204
LRU	0.42515 ± 0.05783	0.31393 ± 0.08024	160,632
Phased LSTM	4.12543 ± 1.05088	0.34936 ± 0.02641	160,614
UnICORNN	0.93700 ± 0.00103	0.93527 ± 0.00084	159,781
Transformer	0.64548 ± 0.06281	0.36811 ± 0.08345	158,860

Table A.2: Learning rates selected by mean validation NRMSE across development seeds 1 and 2. The grid is $\{10^{-4}, 3\times10^{-4}, 10^{-3}\}$; weight decay is 10^{-4} throughout. The selected rate is fixed before final-seed testing.

Model	Oscillator	HalfCheetah	PhysioNet	Δt-irrelevant
WSP	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}
f-CRU	10^{-3}	10^{-3}	10^{-3}	10^{-3}
LRU	10^{-3}	3×10^{-4}	10^{-3}	10^{-3}
Phased LSTM	10^{-3}	10^{-3}	10^{-3}	10^{-3}
UnICORNN	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}
Transformer	3×10^{-4}	10^{-3}	10^{-3}	10^{-3}

Appendix B. Paired statistical comparisons

Table B.3: Paired test comparisons against WSP. Δ is competitor NRMSE minus WSP NRMSE, so positive values favor WSP. Intervals are paired-by-seed percentile bootstrap 95% confidence intervals. Two-sided sign-flip p-values are Holm-adjusted jointly across all 20 comparisons. WSP wins counts seeds with strictly lower WSP error. The bootstrap intervals are not simultaneous intervals.

Competitor	Δ	95% CI	Holm p	WSP wins
Damped oscillator				
f-CRU	0.3058	[0.2623, 0.3463]	0.0391	10/10
LRU	0.1882	[0.1626, 0.2146]	0.0391	10/10
Phased LSTM	1.0828	[0.8927, 1.3051]	0.0391	10/10
UnICORNN	0.6927	[0.6695, 0.7202]	0.0391	10/10
Transformer	0.0229	[−0.0633, 0.1076]	1.0000	5/10
HalfCheetah				
f-CRU	0.5576	[0.5296, 0.5828]	0.0391	10/10
LRU	0.2654	[0.2418, 0.2898]	0.0391	10/10
Phased LSTM	1.9526	[1.6098, 2.3365]	0.0391	10/10
UnICORNN	0.4713	[0.4646, 0.4774]	0.0391	10/10
Transformer	0.3877	[0.3487, 0.4355]	0.0391	10/10
PhysioNet				
f-CRU	0.00050	[−0.00086, 0.00209]	1.0000	6/10
LRU	-0.00309	[−0.00435, −0.00198]	0.0391	0/10
Phased LSTM	0.02489	[0.02356, 0.02624]	0.0391	10/10
UnICORNN	0.06406	[0.06313, 0.06506]	0.0391	10/10
Transformer	0.00055	[−0.00072, 0.00166]	1.0000	7/10
Δt-*irrelevant*				
f-CRU	0.4031	[0.3327, 0.4840]	0.0391	10/10
LRU	0.4083	[0.3726, 0.4405]	0.0391	10/10
Phased LSTM	4.1086	[3.5467, 4.7768]	0.0391	10/10
UnICORNN	0.9201	[0.9194, 0.9209]	0.0391	10/10
Transformer	0.6286	[0.5927, 0.6656]	0.0391	10/10

Appendix C. Elapsed-time interventions and generalization

Table C.4: Test NRMSE by elapsed-time regime (mean ± SD over ten seeds). Interpolation denotes unseen gaps inside the training interval range; extrapolation denotes gaps outside it. Each oscillator evaluation contains 7,292 interpolation and 43,908 extrapolation steps; HalfCheetah contains 13,850 and 82,150 steps, respectively. The extrapolation category therefore constitutes 85.8% and 85.6% of these test steps. These counts describe the common test population, not independent seed replications.

Model	Interpolation	Extrapolation
Damped oscillator		
WSP	0.1134 ± 0.0094	1.9835 ± 0.0187
f-CRU	0.1382 ± 0.0653	2.3134 ± 0.0701
LRU	0.2686 ± 0.0849	2.1843 ± 0.0446
Phased LSTM	1.6839 ± 0.5585	3.0747 ± 0.3330
UnICORNN	0.5285 ± 0.1091	2.7233 ± 0.0404
Transformer	0.2364 ± 0.0473	2.0063 ± 0.1575
HalfCheetah		
WSP	0.2763 ± 0.0074	0.7600 ± 0.0118
f-CRU	0.6678 ± 0.1377	1.3425 ± 0.0479
LRU	0.4309 ± 0.0326	1.0403 ± 0.0456
Phased LSTM	2.4579 ± 0.6472	2.6963 ± 0.6153
UnICORNN	0.6754 ± 0.0004	1.2474 ± 0.0003
Transformer	0.5122 ± 0.0409	1.1686 ± 0.0753

PhysioNet contains 27,452 steps with seen elapsed times, 35 interpolation steps, and one extrapolation step in this benchmark. The single extrapolation transition does not support a population-level claim about real-world long-gap generalization.

Table C.5: WSP test NRMSE under elapsed-time interventions (mean ± SD over ten seeds). Targets and observations are unchanged. Shuffling permutes time gaps within each evaluated batch; constant replaces them with that batch's mean. Noisy uses $\max\{0, \Delta t(1 + 0.2\epsilon)\}$ with $\epsilon \sim \mathcal{N}(0, 1)$. Zero time probes the imposed zero-displacement boundary against unchanged nonzero-time targets, and is not an isolated test of learned timing dependence.

Time input	Test NRMSE
Damped oscillator	
Correct	1.83733 ± 0.01738
Shuffled	1.94028 ± 0.01168
Constant	1.85687 ± 0.02531
Zero	2.47633 ± 0.00000
$0.5\Delta t$	1.89120 ± 0.01279
$2\Delta t$	1.85730 ± 0.02072
Noisy	1.84634 ± 0.01774
HalfCheetah	
Correct	0.71079 ± 0.01100
Shuffled	0.81203 ± 0.00603
Constant	0.75037 ± 0.00554
Zero	1.18265 ± 0.00000
$0.5\Delta t$	0.80168 ± 0.00509
$2\Delta t$	0.72954 ± 0.00828
Noisy	0.71396 ± 0.00944
PhysioNet	
Correct	0.90859 ± 0.00166
Shuffled	0.91448 ± 0.00177
Constant	0.91254 ± 0.00233
Zero	0.97334 ± 0.00000
$0.5\Delta t$	0.91216 ± 0.00201
$2\Delta t$	0.91619 ± 0.00273
Noisy	0.90884 ± 0.00179

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Table C.5 (continued)

Time input	Test NRMSE
Δt-*irrelevant*	
Correct	0.01687 ± 0.00060
Shuffled	0.01679 ± 0.00061
Constant	0.01393 ± 0.00064
Zero	1.00510 ± 0.00000
$0.5\Delta t$	0.02382 ± 0.00100
$2\Delta t$	0.01691 ± 0.00075
Noisy	0.01770 ± 0.00060

Table C.6: Paired WSP shuffled-time effects across ten seeds. The effect is shuffled-time NRMSE minus correct-time NRMSE; positive values indicate deterioration. These 95% bootstrap intervals quantify seed variability, not uncertainty over independently sampled datasets.

Dataset	Shuffling effect	95% CI
Damped oscillator	0.10296	[0.09932, 0.10657]
HalfCheetah	0.10123	[0.09717, 0.10481]
PhysioNet	0.00589	[0.00455, 0.00732]
Δt-irrelevant	-0.00008	[−0.00023, 0.00008]

Appendix D. Autoregressive rollout details

Table D.7: Autoregressive rollout NRMSE (mean ± SD over ten seeds), aggregated over the first H predicted transitions and normalized by training-state standard deviations. This is not endpoint-only error. Predicted states are fed back, while action and time inputs follow the recorded test sequence. These are open-loop prediction evaluations, not closed-loop policy performance. SDs below 0.0005 round to 0.000.

Model	$H = 4$	$H = 8$	$H = 16$	$H = 32$
Damped oscillator				
WSP	0.379 ± 0.006	0.602 ± 0.007	0.867 ± 0.005	0.835 ± 0.003
f-CRU	0.437 ± 0.006	0.643 ± 0.013	0.826 ± 0.027	0.757 ± 0.033
LRU	0.412 ± 0.009	0.653 ± 0.016	0.937 ± 0.025	0.918 ± 0.053
Phased LSTM	0.465 ± 0.017	0.701 ± 0.024	0.897 ± 0.036	1.327 ± 0.271
UnICORNN	0.531 ± 0.000	0.820 ± 0.001	1.006 ± 0.006	0.838 ± 0.011
Transformer	0.374 ± 0.038	0.612 ± 0.054	0.910 ± 0.052	0.955 ± 0.033
HalfCheetah				
WSP	0.626 ± 0.013	0.650 ± 0.022	0.675 ± 0.035	0.705 ± 0.065
f-CRU	1.148 ± 0.091	1.635 ± 0.190	2.513 ± 0.521	3.906 ± 1.189
LRU	0.787 ± 0.033	0.799 ± 0.035	0.811 ± 0.036	0.832 ± 0.037
Phased LSTM	0.872 ± 0.032	1.009 ± 0.061	2.953 ± 0.547	20.063 ± 5.372
UnICORNN	0.830 ± 0.000	0.896 ± 0.000	0.972 ± 0.002	1.321 ± 0.015
Transformer	0.906 ± 0.098	0.942 ± 0.103	0.972 ± 0.105	0.999 ± 0.111
PhysioNet				
WSP	0.944 ± 0.001	0.968 ± 0.003	0.958 ± 0.005	0.933 ± 0.007
f-CRU	0.942 ± 0.001	0.964 ± 0.002	0.951 ± 0.004	0.926 ± 0.011
LRU	0.946 ± 0.001	0.964 ± 0.001	0.948 ± 0.002	0.918 ± 0.004
Phased LSTM	0.951 ± 0.003	0.970 ± 0.002	0.957 ± 0.002	0.937 ± 0.007
UnICORNN	0.980 ± 0.000	0.999 ± 0.000	0.984 ± 0.000	0.948 ± 0.000
Transformer	0.950 ± 0.003	0.967 ± 0.003	0.953 ± 0.004	0.930 ± 0.009

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Table D.7 (continued)

Model	$H = 4$	$H = 8$	$H = 16$	$H = 32$
Δt-*irrelevant*				
WSP	0.008 ± 0.001	0.008 ± 0.001	0.007 ± 0.001	0.006 ± 0.000
f-CRU	0.116 ± 0.024	0.138 ± 0.030	0.154 ± 0.037	0.166 ± 0.044
LRU	0.143 ± 0.038	0.182 ± 0.046	0.217 ± 0.044	0.244 ± 0.041
Phased LSTM	0.263 ± 0.020	0.356 ± 0.026	0.535 ± 0.047	3.167 ± 0.785
UnICORNN	0.451 ± 0.000	0.697 ± 0.000	0.989 ± 0.001	1.298 ± 0.007
Transformer	0.163 ± 0.053	0.222 ± 0.075	0.280 ± 0.083	0.317 ± 0.071

Appendix E. Inference efficiency and training resources

Table E.8: Accuracy-matched PhysioNet inference workloads on one NVIDIA B200. Entries are means across ten checkpoints; each checkpoint uses 30 timed forward passes after warm-up. Throughput counts transitions per second. Memory is peak PyTorch allocated MiB, not GPU-board usage or incremental activation memory. Inputs are shape-matched synthetic benchmark tensors; data loading and application preprocessing are excluded. BF16 autocast is enabled for eligible kernels; f-CRU dynamics remain FP32. Matrix TF32 is disabled.

Model	Validation NRMSE	Transitions/s	Peak MiB
Batch size 1; sequence length 32			
WSP	0.91789	1,023	10.08
f-CRU	0.91493	1,094	11.02
LRU	0.90896	7,979	10.01
Transformer	0.91656	23,694	10.10
Batch size 16; sequence length 32			
WSP	0.91789	15,446	10.34
f-CRU	0.91493	16,040	21.48
LRU	0.90896	122,095	13.02
Transformer	0.91656	343,211	11.17
Batch size 256; sequence length 32			
WSP	0.91789	239,842	14.57
f-CRU	0.91493	257,187	196.34
LRU	0.90896	1,989,429	62.10
Transformer	0.91656	5,488,554	29.69

Table E.9: All nine workload pairs satisfying the 1% validation-accuracy tolerance. The gap is $100|E_{\text{other}} - E_{\text{WSP}}|/E_{\text{WSP}}$ using mean validation NRMSE. The throughput ratio is WSP/competitor, so values below one favor the competitor. Memory saving is $100(1 - M_{\text{WSP}}/M_{\text{other}})$; negative values mean WSP uses more memory. The tolerance is a point-estimate matching rule, not a statistical equivalence test.

Batch	Competitor	Gap (%)	Throughput ratio	Memory saving (%)
1	f-CRU	0.322	0.935	8.54
1	LRU	0.972	0.128	-0.67
1	Transformer	0.144	0.043	0.24
16	f-CRU	0.322	0.963	51.87
16	LRU	0.972	0.127	20.62
16	Transformer	0.144	0.045	7.45
256	f-CRU	0.322	0.933	92.58
256	LRU	0.972	0.121	76.54
256	Transformer	0.144	0.044	50.94

Table E.10: Training-resource measurements across the ten final fits: elapsed training time (minutes, mean ± SD), mean peak allocated training memory (MiB), and mean validation-selected epoch. Training time reflects each fit's stopping point and the parallel training schedule; it is not an isolated, accuracy-matched training-speed benchmark. Training memory includes the training-time working set and must not be equated with inference memory.

Model	Training min	Peak training MiB	Best epoch
Damped oscillator			
WSP	7.1 ± 3.4	242.5	27.1
f-CRU	12.1 ± 4.0	8031.8	57.4
LRU	0.7 ± 0.4	159.7	13.2
Phased LSTM	1.9 ± 0.3	152.4	21.8
UnICORNN	1.1 ± 0.4	165.3	28.4
Transformer	0.3 ± 0.2	86.2	16.1
HalfCheetah			
WSP	36.2 ± 6.9	259.4	69.4
f-CRU	33.6 ± 3.5	8048.5	75.0
LRU	1.0 ± 0.1	176.1	1.7
Phased LSTM	5.6 ± 0.8	166.9	15.0
UnICORNN	9.4 ± 1.3	178.5	90.5
Transformer	0.9 ± 0.2	103.6	1.2
PhysioNet			
WSP	3.5 ± 0.5	249.2	22.6
f-CRU	5.7 ± 0.8	8038.3	43.6
LRU	0.3 ± 0.1	166.0	11.0
Phased LSTM	3.0 ± 0.3	159.2	95.5
UnICORNN	1.5 ± 0.2	170.5	96.9
Transformer	0.4 ± 0.1	92.8	29.8

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Table E.10 (continued)

Model	Training min	Peak training MiB	Best epoch
Δt-*irrelevant*			
WSP	19.3 ± 2.3	247.0	96.1
f-CRU	18.0 ± 1.0	8036.2	98.9
LRU	0.5 ± 0.2	164.1	3.6
Phased LSTM	6.7 ± 0.7	156.8	96.2
UnICORNN	3.4 ± 0.3	168.5	92.1
Transformer	1.0 ± 0.6	90.7	37.1

Appendix F. Independent development-only diagnostics

This development study compares additional time-aware and diagonal state-space baselines under sequence and transition protocols. Each model selects among six combinations of learning rate $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ and weight decay $\{0, 10^{-4}\}$ by mean validation NRMSE over development seeds 1, 2, and 3. The tabulated values are the mean and sample SD for the selected combination. They are development diagnostics, not held-out final-seed test comparisons, and are not pooled with the primary benchmark. PhysioNet uses an independent development preparation; no cross-study pooling of PhysioNet scores is performed. A ranking change between these validation results and the primary test results is not evidence of an inconsistency: the data preparation, split, optimization search, and evaluation protocol must match before quantitative comparisons are valid.

Table F.11: Sequence-protocol development validation NRMSE (mean ± SD over three seeds) with selected learning rate (LR) and weight decay (WD). These values characterize validation-based model selection and are not test-set performance estimates.

Model	Validation NRMSE	LR	WD
Damped oscillator			
WSP	0.1701 ± 0.0144	10^{-3}	0
Conditioned diagonal SSM	0.1517 ± 0.0059	10^{-3}	10^{-4}
Global diagonal SSM	0.1269 ± 0.0046	10^{-3}	10^{-4}
TGNN4I probe	0.0706 ± 0.0074	10^{-3}	10^{-4}
Time-GRU	0.3096 ± 0.0710	10^{-3}	0
Time-MLP	0.1956 ± 0.0165	10^{-4}	0
Transformer	0.4216 ± 0.0574	10^{-3}	10^{-4}

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Table F.11 (continued)

Model	Validation NRMSE	LR	WD
HalfCheetah			
WSP	0.2448 ± 0.0129	10^{-3}	0
Conditioned diagonal SSM	0.2681 ± 0.0022	10^{-3}	10^{-4}
Global diagonal SSM	0.3486 ± 0.0008	10^{-3}	0
TGNN4I probe	0.2297 ± 0.0091	10^{-3}	10^{-4}
Time-GRU	0.4092 ± 0.0092	3×10^{-4}	10^{-4}
Time-MLP	0.3917 ± 0.0251	10^{-4}	0
Transformer	0.5128 ± 0.0569	10^{-3}	10^{-4}
Hopper			
WSP	0.2179 ± 0.0167	10^{-3}	0
Conditioned diagonal SSM	0.2579 ± 0.0002	10^{-3}	10^{-4}
Global diagonal SSM	0.3624 ± 0.0010	10^{-3}	0
TGNN4I probe	0.1632 ± 0.0035	10^{-3}	0
Time-GRU	0.3954 ± 0.0295	3×10^{-4}	10^{-4}
Time-MLP	0.3297 ± 0.0107	10^{-3}	0
Transformer	0.4890 ± 0.0403	10^{-3}	0
Reacher			
WSP	0.2109 ± 0.0102	10^{-3}	10^{-4}
Conditioned diagonal SSM	0.2042 ± 0.0047	10^{-3}	0
Global diagonal SSM	0.4815 ± 0.0017	10^{-3}	10^{-4}
TGNN4I probe	0.2207 ± 0.0028	10^{-3}	10^{-4}
Time-GRU	0.5267 ± 0.0295	10^{-3}	10^{-4}
Time-MLP	0.4564 ± 0.0328	10^{-3}	0
Transformer	0.5633 ± 0.0292	10^{-3}	0
Δt-*irrelevant*			
WSP	0.0252 ± 0.0003	10^{-3}	0
Conditioned diagonal SSM	0.0530 ± 0.0029	10^{-3}	0
Global diagonal SSM	0.0583 ± 0.0001	10^{-3}	0
TGNN4I probe	0.0610 ± 0.0108	10^{-3}	0
Time-GRU	0.2172 ± 0.0350	3×10^{-4}	10^{-4}
Time-MLP	0.1856 ± 0.0191	10^{-4}	0
Transformer	0.3735 ± 0.1677	10^{-3}	0

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Table F.11 (continued)

Model	Validation NRMSE	LR	WD
PhysioNet			
WSP	0.9197 ± 0.0006	10^{-3}	10^{-4}
Conditioned diagonal SSM	0.9176 ± 0.0005	10^{-3}	10^{-4}
Global diagonal SSM	0.9231 ± 0.0001	10^{-3}	10^{-4}
TGNN4I probe	0.9142 ± 0.0001	10^{-3}	10^{-4}
Time-GRU	0.9087 ± 0.0004	10^{-3}	10^{-4}
Time-MLP	0.9338 ± 0.0004	10^{-3}	0
Transformer	0.9163 ± 0.0007	10^{-3}	0

Table F.12: Transition-protocol development validation NRMSE (mean ± SD over three seeds) with selected learning rate (LR) and weight decay (WD). These values characterize validation-based model selection and are not test-set performance estimates.

Model	Validation NRMSE	LR	WD
Damped oscillator			
WSP (local transition)	0.0151 ± 0.0013	10^{-4}	0
Conditioned diagonal SSM	0.0097 ± 0.0012	10^{-3}	0
Global diagonal SSM	0.0104 ± 0.0019	3×10^{-4}	10^{-4}
TGNN4I probe	0.0136 ± 0.0007	3×10^{-4}	10^{-4}
Time-MLP	0.1767 ± 0.1062	10^{-4}	0
Transformer	0.1359 ± 0.0102	10^{-3}	0
Feature-token Transformer	0.0148 ± 0.0019	10^{-4}	0
HalfCheetah			
WSP (local transition)	0.0896 ± 0.0009	10^{-3}	0
Conditioned diagonal SSM	0.0955 ± 0.0063	10^{-3}	0
Global diagonal SSM	0.2130 ± 0.0024	10^{-3}	0
TGNN4I probe	0.1268 ± 0.0009	10^{-3}	0
Time-MLP	0.6039 ± 0.0460	10^{-4}	0
Transformer	0.3356 ± 0.0308	10^{-4}	0
Feature-token Transformer	0.1206 ± 0.0228	3×10^{-4}	0

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Table F.12 (continued)

Model	Validation NRMSE	LR	WD
Hopper			
WSP (local transition)	0.0696 ± 0.0007	10^{-3}	10^{-4}
Conditioned diagonal SSM	0.0676 ± 0.0035	10^{-3}	0
Global diagonal SSM	0.2059 ± 0.0070	10^{-3}	10^{-4}
TGNN4I probe	0.0744 ± 0.0016	10^{-3}	0
Time-MLP	0.3947 ± 0.0428	10^{-3}	0
Transformer	0.2149 ± 0.0132	10^{-3}	10^{-4}
Feature-token Transformer	0.0715 ± 0.0125	10^{-3}	10^{-4}
Reacher			
WSP (local transition)	0.0666 ± 0.0012	10^{-4}	0
Conditioned diagonal SSM	0.0643 ± 0.0012	3×10^{-4}	10^{-4}
Global diagonal SSM	0.2878 ± 0.0730	10^{-3}	10^{-4}
TGNN4I probe	0.0789 ± 0.0058	10^{-3}	0
Time-MLP	0.5580 ± 0.0637	3×10^{-4}	10^{-4}
Transformer	0.2988 ± 0.0089	10^{-4}	0
Feature-token Transformer	0.0907 ± 0.0195	10^{-3}	10^{-4}
Δt-*irrelevant*			
WSP (local transition)	0.0044 ± 0.0003	10^{-3}	10^{-4}
Conditioned diagonal SSM	0.0056 ± 0.0001	10^{-3}	0
Global diagonal SSM	0.0076 ± 0.0002	10^{-3}	0
TGNN4I probe	0.0077 ± 0.0005	10^{-3}	10^{-4}
Time-MLP	0.0818 ± 0.0233	10^{-4}	0
Transformer	0.2118 ± 0.0463	10^{-3}	10^{-4}
Feature-token Transformer	0.0066 ± 0.0012	3×10^{-4}	10^{-4}

Appendix G. Independent Controlled Component Study

This five-seed study isolates changes within the persistent WSP architecture. It uses its own training configuration and dataset preparation and is not pooled with the ten-seed primary benchmark. Reacher, HalfCheetah, and the damped oscillator use 32-transition windows with stride 16; Hopper uses four-transition windows with stride 2. WSP component variants retain the same parameter tensors while disabling the indicated computation. The smaller pointwise WFB and complex SSM controls are diagnostic rather than parameter-matched competitors.

WSP uses width 128, two layers, rank 16, and seeds 1–5. Training uses batch size 256, learning rate 3×10^{-4}, weight decay 10^{-4}, at most 75 epochs, early-stopping patience 15, gradient clipping at 1, and BF16 mixed precision. These settings are specific to this study, not the primary benchmark's 100-epoch protocol.

Carry-minus-reset claims refer only to this controlled study. The paired 95% Student-t intervals for reset-minus-carry NRMSE, based on five seeds, are $[0.0822, 0.1168]$ for Reacher, $[0.4028, 0.5266]$ for HalfCheetah, $[-0.0057, 0.0860]$ for the oscillator, and $[0.0352, 0.1234]$ for four-step Hopper. These exploratory intervals are not multiplicity-adjusted.

Table G.13: Ordered-window control comparison. Test NRMSE is the mean ± standard deviation over five seeds. The final column reports exact trainable parameter counts. Reacher, HalfCheetah, and the oscillator use length 32; Hopper uses length 4. The pointwise WFB and complex SSM are smaller diagnostic controls rather than capacity-matched baselines.

Model	Test NRMSE	Parameters
Reacher (length 32)		
WSP carry	**1.2372 ± .0086**	161,152
Time MLP	1.2877 ± .0135	160,522
Time GRU	1.3141 ± .0230	159,850
Transformer	1.4471 ± .0450	173,986
Sine MLP	2.0932 ± .0656	163,474
Pointwise WFB	1.8777 ± .0377	13,834
Complex SSM	1.3505 ± .0061	24,576
HalfCheetah (length 32)		
WSP carry	**.6994 ± .0047**	163,456
Time MLP	.8927 ± .0161	160,945
Time GRU	.8564 ± .0096	162,737
Transformer	.9985 ± .0602	175,577
Sine MLP	1.7160 ± .1243	163,985
Pointwise WFB	1.4781 ± .0295	23,057
Complex SSM	.8514 ± .0003	43,008
Damped oscillator (length 32)		
WSP carry	**1.6716 ± .0077**	158,976
Time MLP	1.6743 ± .0196	160,330
Time GRU	1.7756 ± .0278	157,122
Transformer	1.9519 ± .0432	146,562
Sine MLP	2.2456 ± .0938	156,802
Pointwise WFB	2.3307 ± .0187	5,122
Complex SSM	1.8589 ± .0073	7,168

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Table G.13 (continued)

Model	Test NRMSE	Parameters
Hopper (length 4)		
WSP carry	**3.0315 ± .0304**	161,536
Time MLP	3.1273 ± .0386	161,675
Time GRU	3.2713 ± .0173	160,331
Transformer	3.4554 ± .0314	174,251
Sine MLP	3.5706 ± .0116	164,651
Pointwise WFB	3.8894 ± .0099	15,371
Complex SSM	3.4493 ± .0085	27,648

Table G.14: Sequence-level test NRMSE (mean ± standard deviation over five seeds; lower is better). Carry propagates the wave state between transitions.

Environment	Carry	Reset	No damping	No frequency
Reacher	1.2372 ± .0086	1.3367 ± .0158	1.5912 ± .0427	1.2932 ± .0030
HalfCheetah	.6994 ± .0047	1.1641 ± .0496	1.1446 ± .1017	.7002 ± .0060
Damped oscillator	1.6716 ± .0077	1.7117 ± .0355	1.8756 ± .0826	1.6264 ± .0230
Hopper (length 4)	3.0315 ± .0304	3.1108 ± .0071	3.2172 ± .0188	**2.7633 ± .0091**

Environment	Cosine only	No conditioner	No zero-time
Reacher	1.3361 ± .0247	1.2571 ± .0046	**1.2211 ± .0043**
HalfCheetah	.9353 ± .0243	**.6849 ± .0031**	.6968 ± .0087
Damped oscillator	1.8046 ± .0233	**1.6212 ± .0087**	1.6708 ± .0130
Hopper (length 4)	3.0039 ± .0562	3.2311 ± .0175	3.0086 ± .0167

Table G.15: Relative NRMSE increase after elapsed-time interventions on WSP carry. Positive values indicate degradation from the correct intervals.

Environment	Shuffled	Reversed	Zero
Reacher	6.5%	6.8%	32.5%
HalfCheetah	14.5%	15.1%	68.0%
Damped oscillator	5.4%	5.7%	35.1%
Hopper (length 4)	5.5%	6.1%	27.2%

Appendix H. Learning-History Diagnostics

The following summaries use the recorded histories of the ten final HalfCheetah seeds for each model. They diagnose the declared training protocol without changing checkpoint selection or evaluating new models. Online training losses and post-epoch validation errors are distinct quantities; early-stopping epochs may differ across seeds.

Table H.16: HalfCheetah learning-history audit over ten final-run seeds. First denotes epoch 1, best the validation-selected epoch, and last the actual stopping epoch. Validation NRMSE is mean ± sample standard deviation. A displayed SD of 0.000 denotes a positive value below 0.0005. Epoch ranges span the ten seeds; runs stop after 20 non-improving epochs or at epoch 100.

Model	Best epoch	Last epoch	First val.	Best val.	Last val.
WSP	38–88	58–100	0.365 ± 0.003	0.235 ± 0.009	0.237 ± 0.009
f-CRU	48–90	68–100	0.858 ± 0.011	0.228 ± 0.010	0.230 ± 0.011
LRU	1–2	21–22	0.550 ± 0.037	0.537 ± 0.044	0.853 ± 0.100
Phased LSTM	10–19	30–39	0.974 ± 0.000	0.835 ± 0.021	1.134 ± 0.124
UnICORNN	84–97	100	0.975 ± 0.000	0.972 ± 0.000	0.972 ± 0.000
Transformer	1–2	21–22	0.579 ± 0.025	0.576 ± 0.028	0.869 ± 0.064

Table H.17: HalfCheetah online training losses for the same ten runs (mean ± sample standard deviation). Training MSE is accumulated while parameters change during an epoch; it is not a post-epoch checkpoint evaluation. Reduction is $100(1 - M_{\text{last}}/M_1)$ computed within each seed and then averaged. Displayed SDs of 0.0000 or 0.0 denote positive values below half a unit in the final shown digit. Validation and training use their original, distinct units.

Model	First train MSE	Last train MSE	Reduction (%)
WSP	0.3394 ± 0.0034	0.0177 ± 0.0009	94.8 ± 0.3
f-CRU	0.9275 ± 0.0053	0.0341 ± 0.0008	96.3 ± 0.1
LRU	0.5149 ± 0.0105	0.0454 ± 0.0009	91.2 ± 0.2
Phased LSTM	0.9987 ± 0.0002	0.4132 ± 0.0301	58.6 ± 3.0
UnICORNN	0.9991 ± 0.0000	0.9922 ± 0.0003	0.7 ± 0.0
Transformer	0.6202 ± 0.0020	0.0293 ± 0.0004	95.3 ± 0.1

Appendix I. Selected-Validation Diagonal-SSM Contrasts

These descriptive contrasts use three development seeds and the settings selected from a six-configuration learning-rate–weight-decay grid. All selected WSP and diagonal-SSM settings use learning rate 10^{-3}, the upper edge of that grid. Thus the comparison does not demonstrate optimization saturation. Hopper uses four-transition windows; the other datasets use 32. The PhysioNet development preparation is independent of the primary comparator benchmark. Exact absolute validation NRMSE, learning rates, and weight decay are in Table F.11; no final test values are substituted for these development quantities.

Table I.18: Selected-validation contrasts with diagonal-SSM controls. Relative reduction is $100(\bar{e}_{\mathrm{SSM}} - \bar{e}_{\mathrm{WSP}})/\bar{e}_{\mathrm{SSM}}$. Positive values favor WSP. The range spans the three paired development-seed reductions and is not a confidence interval. The final column counts seeds with lower WSP validation NRMSE. All configurations and checkpoints were chosen using these validation data. Hopper uses length 4; other datasets use length 32. PhysioNet follows the separate development preparation.

Comparator	Reduction (%)	Seed range (%)	WSP lower
Damped oscillator			
Global diagonal SSM	−34.02	[−46.77, −17.14]	0/3
Conditioned diagonal SSM	−12.13	[−22.29, −5.44]	0/3
HalfCheetah			
Global diagonal SSM	+29.77	[+25.97, +33.65]	3/3
Conditioned diagonal SSM	+8.66	[+3.03, +13.66]	3/3
Hopper ($L = 4$)			
Global diagonal SSM	+39.87	[+34.67, +43.64]	3/3
Conditioned diagonal SSM	+15.48	[+8.42, +20.86]	3/3
Reacher			
Global diagonal SSM	+56.20	[+54.63, +58.65]	3/3
Conditioned diagonal SSM	−3.30	[−10.11, +3.08]	1/3

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Table I.18 (continued)

Comparator	Reduction (%)	Seed range (%)	WSP lower
Δt-irrelevant			
Global diagonal SSM	+56.81	[+56.49, +57.31]	3/3
Conditioned diagonal SSM	+52.53	[+50.19, +55.06]	3/3
PhysioNet			
Global diagonal SSM	+0.37	[+0.28, +0.42]	3/3
Conditioned diagonal SSM	−0.23	[−0.34, −0.15]	0/3

Appendix J. Saved Prefix and Interval Diagnostics

The prefix metrics score the first L targets of each fixed 32-transition test window after resetting the model state at the window boundary. They do not hold the target set fixed while changing the available history. All entries below summarize ten final seeds; the counts are target coordinates per seed, not independent statistical replicates.

Table J.19: Prefix displacement NRMSE: each cell shows the mean with sample SD in parentheses below it. $L = 32$ reproduces the primary test score. Each prefix is scored on its first L targets, not on a fixed target set with truncated history. Values are rounded to four decimals; a displayed zero SD may reflect rounding.

Model	$L = 1$	$L = 4$	$L = 8$	$L = 16$	$L = 32$
Damped oscillator					
WSP	2.2948	2.2898	2.2276	2.0846	1.8373
	(0.0420)	(0.0291)	(0.0240)	(0.0209)	(0.0174)
f-CRU	2.9166	2.8389	2.7166	2.4806	2.1431
	(0.0302)	(0.0281)	(0.0369)	(0.0514)	(0.0643)
LRU	2.5073	2.4922	2.4312	2.2840	2.0255
	(0.0359)	(0.0389)	(0.0408)	(0.0411)	(0.0421)
Phased LSTM	2.8103	2.7776	2.6694	2.6029	2.9202
	(0.0972)	(0.0761)	(0.0739)	(0.0616)	(0.3486)
UnICORNN	3.0833	3.0683	2.9845	2.8140	2.5301
	(0.0001)	(0.0008)	(0.0033)	(0.0081)	(0.0408)
Transformer	2.1276	2.2344	2.1890	2.0798	1.8602
	(0.2349)	(0.2082)	(0.1971)	(0.1764)	(0.1459)

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Table J.19 (continued)

Model	$L = 1$	$L = 4$	$L = 8$	$L = 16$	$L = 32$
HalfCheetah					
WSP	0.7742	0.7257	0.7190	0.7152	0.7108
	(0.0138)	(0.0118)	(0.0113)	(0.0110)	(0.0110)
f-CRU	1.1305	1.1143	1.1700	1.2409	1.2684
	(0.0303)	(0.0334)	(0.0389)	(0.0403)	(0.0471)
LRU	1.0208	0.9793	0.9738	0.9742	0.9762
	(0.0393)	(0.0415)	(0.0413)	(0.0425)	(0.0428)
Phased LSTM	1.1851	1.1883	1.2035	1.4927	2.6634
	(0.0104)	(0.0223)	(0.0360)	(0.1620)	(0.6193)
UnICORNN	1.2214	1.1965	1.1900	1.1852	1.1821
	(0.0000)	(0.0000)	(0.0000)	(0.0001)	(0.0003)
Transformer	1.0216	1.0825	1.0929	1.0978	1.0985
	(0.0348)	(0.0606)	(0.0653)	(0.0680)	(0.0693)
PhysioNet					
WSP	1.5712	1.2893	1.1396	1.0156	0.9086
	(0.0018)	(0.0021)	(0.0021)	(0.0021)	(0.0017)
f-CRU	1.5673	1.2935	1.1478	1.0215	0.9091
	(0.0011)	(0.0008)	(0.0016)	(0.0028)	(0.0019)
LRU	1.5729	1.2895	1.1397	1.0133	0.9055
	(0.0012)	(0.0011)	(0.0011)	(0.0013)	(0.0018)
Phased LSTM	1.5881	1.3244	1.1701	1.0392	0.9335
	(0.0076)	(0.0050)	(0.0039)	(0.0034)	(0.0024)
UnICORNN	1.6295	1.3647	1.2106	1.0788	0.9726
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Transformer	1.5729	1.2968	1.1444	1.0162	0.9091
	(0.0022)	(0.0025)	(0.0014)	(0.0015)	(0.0016)

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Table J.19 (continued)

Model	$L = 1$	$L = 4$	$L = 8$	$L = 16$	$L = 32$
Δt-irrelevant					
WSP	0.0563	0.0368	0.0284	0.0216	0.0169
	(0.0037)	(0.0018)	(0.0013)	(0.0009)	(0.0006)
f-CRU	0.6409	0.5893	0.5230	0.4614	0.4199
	(0.1695)	(0.1455)	(0.1322)	(0.1276)	(0.1285)
LRU	0.6643	0.5423	0.4840	0.4456	0.4251
	(0.1650)	(0.1062)	(0.0795)	(0.0643)	(0.0578)
Phased LSTM	1.1679	0.8513	0.6767	1.1307	4.1254
	(0.1037)	(0.0670)	(0.0530)	(0.2286)	(1.0509)
UnICORNN	1.6216	1.3893	1.2252	1.0552	0.9370
	(0.0000)	(0.0001)	(0.0002)	(0.0005)	(0.0010)
Transformer	0.7442	0.7404	0.6993	0.6649	0.6455
	(0.1786)	(0.0870)	(0.0554)	(0.0541)	(0.0628)

For each prefix length L, the number of scored target coordinates per seed is 3,200L (Damped oscillator), 51,000L (HalfCheetah), 8,590L (PhysioNet), 9,600L (Δt-irrelevant).

Relative reductions use $100(\bar{e}_b - \bar{e}_{\mathrm{WSP}})/\bar{e}_b$. Intervals are pointwise, unadjusted paired bootstraps (20,000 resamples of ten seeds), not confirmatory tests. Checkpoints were fixed before these interval-specific test contrasts.

Table J.20: WSP relative error reduction by interval. Positive values favor WSP. Steps count transitions per seed; wins count seeds with strictly lower WSP NRMSE.

Comparator	$\Delta t/T$	Steps	Reduction [95% CI], %	Wins
Damped oscillator				
LRU	0.375	7,292	57.78 [49.09, 64.53]	10/10
Transformer	0.375	7,292	52.03 [44.56, 58.06]	10/10
LRU	1.25	7,278	12.47 [−2.02, 24.06]	8/10
Transformer	1.25	7,278	31.78 [25.78, 36.79]	10/10
LRU	1.5	7,390	11.36 [3.07, 17.77]	8/10
Transformer	1.5	7,390	24.87 [19.15, 29.47]	10/10
LRU	2	7,277	25.53 [21.64, 29.01]	10/10
Transformer	2	7,277	31.79 [28.12, 34.75]	10/10
LRU	3	7,279	7.37 [5.87, 8.69]	10/10
Transformer	3	7,279	0.55 [−6.28, 6.35]	4/10
LRU	4	7,371	11.55 [10.26, 12.80]	10/10
Transformer	4	7,371	3.93 [−0.22, 7.56]	8/10
LRU	5	7,313	6.53 [5.44, 7.45]	10/10
Transformer	5	7,313	−6.80 [−11.52, −2.50]	2/10
HalfCheetah				
LRU	0.375	13,850	35.86 [33.50, 38.63]	10/10
Transformer	0.375	13,850	46.05 [43.13, 48.78]	10/10
LRU	1.25	13,670	47.87 [44.06, 51.09]	10/10
Transformer	1.25	13,670	50.90 [47.39, 54.28]	10/10
LRU	1.5	13,801	32.15 [28.90, 35.21]	10/10
Transformer	1.5	13,801	36.75 [33.87, 39.54]	10/10
LRU	2	13,500	30.00 [29.07, 30.87]	10/10
Transformer	2	13,500	30.06 [28.84, 31.01]	10/10
LRU	3	13,631	17.45 [14.88, 20.22]	10/10
Transformer	3	13,631	30.87 [27.79, 34.39]	10/10
LRU	4	13,792	24.52 [22.25, 27.08]	10/10
Transformer	4	13,792	26.23 [23.53, 29.42]	10/10
LRU	5	13,756	21.87 [18.77, 24.89]	10/10
Transformer	5	13,756	40.04 [35.86, 45.04]	10/10

References

[1] Z. Che, S. Purushotham, K. Cho, D. Sontag, Y. Liu, Recurrent neural networks for multivariate time series with missing values, Scientific Reports 8 (2018) 6085. doi:10.1038/s41598-018-24271-9.

[2] Y. Rubanova, R. T. Q. Chen, D. Duvenaud, Latent ordinary differential equations for irregularly-sampled time series, in: Advances in Neural Information Processing Systems, volume 32, 2019.

[3] P. Kidger, J. Morrill, J. Foster, T. Lyons, Neural controlled differential equations for irregular time series, in: Advances in Neural Information Processing Systems, volume 33, 2020, pp. 6696–6707.

[4] Y. Chen, K. Ren, Y. Wang, Y. Fang, W. Sun, D. Li, Contiformer: Continuous-time transformer for irregular time series modeling, in: Advances in Neural Information Processing Systems, volume 36, 2023.

[5] A. Gu, K. Goel, C. Ré, Efficiently modeling long sequences with structured state spaces, in: International Conference on Learning Representations, 2022. URL: https://openreview.net/forum?id=uYLFoz1vlAC.

[6] A. Gupta, A. Gu, J. Berant, On the parameterization and initialization of diagonal state space models, in: Advances in Neural Information Processing Systems, volume 35, 2022, pp. 35971–35983.

[7] J. T. H. Smith, A. Warrington, S. W. Linderman, Simplified state space layers for sequence modeling, in: International Conference on Learning Representations, 2023. URL: https://openreview.net/forum?id=Ai8Hw3AXqks.

[8] A. Orvieto, S. L. Smith, A. Gu, A. Fernando, C. Gulcehre, R. Pascanu, S. De, Resurrecting recurrent neural networks for long sequences, in: Proceedings of the 40th International Conference on Machine Learning, volume 202 of *Proceedings of Machine Learning Research*, 2023, pp. 26670–26698.

[9] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, in: Proceedings of the First Conference on Language Modeling, 2024. URL: https://openreview.net/forum?id=tEYskw1VY2.

[10] B. Yu, J. Kim, Wave function backpropagation with explicit temporal-interval dynamics, 2026. URL: https://arxiv.org/abs/2609.00503, the 38th IEEE International Conference on Tools with Artificial Intelligence (ICTAI).

[11] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, D. Duvenaud, Neural ordinary differential equations, in: Advances in Neural Information Processing Systems, volume 31, 2018.

[12] E. De Brouwer, J. Simm, A. Arany, Y. Moreau, GRU-ODE-Bayes: Continuous modeling of sporadically-observed time series, in: Advances in Neural Information Processing Systems, volume 32, 2019.

[13] R. Hasani, M. Lechner, A. Amini, D. Rus, R. Grosu, Liquid time-constant networks, in: Proceedings of the AAAI Conference on Artificial Intelligence, volume 35, 2021, pp. 7657–7666. doi:10.1609/aaai.v35i9.16936.

[14] D. Neil, M. Pfeiffer, S.-C. Liu, Phased LSTM: Accelerating recurrent network training for long or event-based sequences, in: Advances in Neural Information Processing Systems, volume 29, 2016.

[15] M. Schirmer, M. Eltayeb, S. Lessmann, M. Rudolph, Modeling irregular time series with continuous recurrent units, in: Proceedings of the 39th International Conference on Machine Learning, volume 162 of *Proceedings of Machine Learning Research*, 2022, pp. 19388–19405. URL: https://proceedings.mlr.press/v162/schirmer22a.html.

[16] T. K. Rusch, S. Mishra, UnICORNN: A recurrent model for learning very long time dependencies, in: Proceedings of the 38th International Conference on Machine Learning, volume 139 of *Proceedings of Machine Learning Research*, 2021, pp. 9168–9178. URL: https://proceedings.mlr.press/v139/rusch21a.html.

[17] B. Lusch, J. N. Kutz, S. L. Brunton, Deep learning for universal linear embeddings of nonlinear dynamics, Nature Communications 9 (2018) 4950. doi:10.1038/s41467-018-07210-0.

[18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, volume 30, 2017.

[19] N. Hansen, H. Su, X. Wang, TD-MPC2: Scalable, robust world models for continuous control, in: International Conference on Learning Representations, 2024. URL: https://openreview.net/forum?id=Oxh5CstDJU.

[20] A. Scannell, M. Nakhaeinezhadfard, K. Kujanpää, Y. Zhao, K. S. Luck, A. Solin, J. Pajarinen, Discrete codebook world models for continuous control, in: International Conference on Learning Representations, 2025.

[21] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, G. Wetzstein, Implicit neural representations with periodic activation functions, in: Advances in Neural Information Processing Systems, volume 33, 2020, pp. 7462–7473.

[22] E. Todorov, T. Erez, Y. Tassa, MuJoCo: A physics engine for model-based control, in: 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2012, pp. 5026–5033. doi:10.1109/IROS.2012.6386109.

[23] I. Silva, G. Moody, R. Mark, L. A. Celi, Predicting mortality of ICU patients: The PhysioNet/Computing in Cardiology Challenge 2012, 2012. URL: https://physionet.org/content/challenge-2012/1.0.0/, physioNet, version 1.0.0.

[24] T. Pollard, B. E. Moody, L.-w. H. Lehman, B. J. Gow, C. Fernandes, C. Xie, A. Johnson, R. G. Mark, T. Heldt, PhysioNet as a global platform for biomedical research, Nature Health 1 (2026) 792–795. doi:10.1038/s44360-026-00096-z.